**MILITARY ENGINEER SERVICES**
**COMMANDER WORKS ENGINEER (AF) BAGDOGRA**

**NAME OF WORK: PROVN OF WIDENING OF ROAD OF CERTAIN INTERNAL ROAD AND ALLIED WORKS AT AF STN BAGDOGRA UNDER GE (AF) BAGDOGRA.**

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**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

TELE: 0353-2698307

HQ Commander Works Engineer
(AF) Bagdogra
Post : Bagdogra Airport
Distt : Darjeeling (WB)
Pin : 734 421

**8549/BAG/2023-24/ ____/E8** **26 Aug 2025**

**M/S _____**
**_____**
**_____**
**_____**

**NAME OF WORK: PROVN OF WIDENING OF ROAD OF CERTAIN INTERNAL ROAD AND ALLIED WORKS AT AF STN BAGDOGRA UNDER GE (AF) BAGDOGRA.**

Dear Sir(s),

1. Tender documents in respect of above work are uploaded on the site www.defproc.gov.in.The tender is on single stage two cover e-tendering system. The contents of Cover1 & Cover 2 are specified in NOTICE OF TENDER.

   "Cover 1" shall contain application with complete supporting documents /affidavit required for satisfying the eligibility criteria as specified in NIT including proof for cost of tender and EMD as applicable excluding the Quotation/price bid (BOQ). **"Bidder shall upload copy of GST Registration Number which is mandatory. Bidders not in possession of GST Registration Number shall be disqualified in "Technical Bid" evaluation and their finance bid shall not be opened"**. Bidder shall produce provident fund code Number to GE while signing work order.

   "Cover 2" shall contain only "Bill of Quantity (BOQ) duly quoted by the bidders" i.e. (Schedule duly quoted). The bids of established eligible contractors only shall be opened in the process on the date specified for opening of commercial bids i.e. Cover 2.

2. Bid will be received online by ACCEPTING OFFICER up to the date and time mentioned in the NOTICE INVITING TENDER (NIT). No tender / bid will be received in physical form and any tender/bid received in such manner will be treated as non bonafide tender/bid.

3. Bids will be opened on due date and time fixed for opening in the presence of tenders/bidders or their authorized representatives, who have uploaded their bid and who wish to be present at the time of opening the bids.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents and / or depute your technical representative for discussion on tender/drawings and to clarify doubts, if any, on or before clarification end date. You are not to write piece meal points and forward your points duly consolidated before clarification end date.

5. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appendix-'A' to NIT on e-procurement portal and submit physical documents in the office of the "HQ COMMANDER WORKS ENGINEER (AF) BAGDOGRA, POST: BAGDOGRA AIRPORT, DISTT: DARJEELING (WB), PIN: 734 421" before date & time fixed for this purpose.

6. The contractor must ensure that the tender/bid on the proper form is uploaded in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand /through post from tenderer/bidder even if they are received in time.

7. In view of delays due to system failure or other communication related failures, it is suggested that the tender/bid be uploaded, if necessary, sufficient in advance of the last due date and time fixed.

8. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of minimum fair wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in the Office of GE concerned and this office.

9. ANY TENDERER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITION, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED.

10. This letter shall form part of the tender documents.

**Yours faithfully,**

**SIGNATURE OF CONTRACTOR**
**DATED: _____**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

## INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. **EARNEST MONEY DEPOSIT (EMD):**

All bidders shall be exempted from submission of EMD subject to submission of signed **"Bid Securing Declaration"** in Cover-1 which is as under:-

(i) I/We hereby understand and accept that if I/We withdraw or modify my/our bids during the period of validity, or if I/We are awarded the contracts and on being called upon to submit the Performance Security/Security Deposit, fail to submit the Performance Security/Security Deposit before the deadline defined in the request for bid documents/Notice Inviting Tender, I/We shall be debarred from exemption of submitting Bid Security/Earnest Money Deposit for a period of 6 (Six) Months, from the date I/We are declared disqualified from exemption from submission of Earnest Money Deposit / Security Deposit, for all tender issued by MES during this period.

(ii) I/we hereby certify that there are no outstanding recoveries against our firm and I/we do not intend to abandon the work after entertaining the relief.

2. **PERFORMANCE SECURITY DEPOSIT (APPLICABLE FOR BOTH UNENLISTED AS WELL AS ENLISTED CONTRACTORS):**

In case, the tender submitted by contractor is accepted, the contractor will be required to lodge "**Performance Security Deposit"** for an amount equivalent to **5 % of the Contract Sum/lumpsum** in favour of the Accepting Officer in the form of "**Bank Guarantee"** or **FDR.** Work Order No.01 shall be placed only after submission of Performance Security of adequate value by the Contractor. In case a fixed deposit receipt of any Bank is furnished by the contractor to the Government as part of the Performance Guarantee and the Bank is unable to make payment against the said fixed deposit receipt, the loss caused thereby shall fall on the contractor and the contractor shall forthwith on demand furnish additional security to the Government to make good the deficit. The period of validity of the Bank Guarantee Bond against Performance Security shall be initially valid up to the stipulated date of expiry of Defects Liability Period plus minimum 60 days beyond that. In case final bill not paid during this period, the contractor shall get the validity of Performance Guarantee extended to cover such enlarged time required for payment of final bill.

3. **GENERAL INSTRUCTIONS FOR COMPLIANCE**:

(a) The bids received only in the electronic form will be considered. All bids shall be submitted on defproc.gov.in portal. Documents should be scanned and forwarded in "pdf" form and "xls" form as indicated.

(b) Bids shall be uploaded on defproc.gov.in portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like email/fax/by hand/through post will be considered.

(c) Bid should be DIGITALLY signed using valid DSC. All pages of tender documents, corrections/alterations shall be signed/initialed by the lowest bidder after acceptance.

(d) Drawings, if issued in physical form, must be returned duly initialed by the tender/bidder in separate envelope indicating his name and address.

(e) The tender shall be signed, dated and witness at all places provided for in the documents after acceptance. All corrections shall be initialed. The Contractor shall initial every pages of tender and shall sign all drawings forming part of tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

**INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD…)**

(f) In the technical bid, scanned copy of Power attorney in favour of the person uploading the bid using his/her DSC shall be uploaded. In case, the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clause, shall be attached in "pdf" form. In case of partnership concern or a limited company, digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Articles of Association of the Company) in all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the documents confirming of the authority shall be attached with the tender/bid in "pdf" form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of a firm or company using his DSC shall uploaded with the tender/bid a scanned copy (in "pdf" form) of Power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be, in all matters pertaining to the contract including the Arbitration Clause.

(g) Even in case of Firm or Companies which have already given Power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh Power of Attorney duly executed in his favour stating specifically that the said person has authority to bind such partners of the Firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in "pdf" form with the tender/bid; unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Articles of Association.

(h) Hard copies of all above documents should be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

(j) Bid (Cover 1 & 2) shall be uploaded online well in time.

(k) The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to special condition 3 referred hereinafter and also conditions 24& 25 of IAFW 2249(General conditions of contract).

(l) Tenderers/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tender/bids, may do so at the appointed time.

(m) The tenderer/bidder shall quote his rate on the BOQ file only. No alteration to the format will be accepted; else the bid will be disqualified and summarily rejected.

(n) In case the tenderer/bidder has to revise/modify the rates quoted in the BOQ (excel sheet) he can do so only in the BOQ, through defproc.gov.in in site only before the bid closing time and date.

**INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD…)**

4. **REVOCATION/REVISION OF OFFER UPWARD/OFFERING VOLUNTARY REDUCTION, AFTER OPENING OF FINANCIAL BIDS BY LOWEST BIDDER:**
In the event of lowest tenderer/bidder revoking his offer or revising his rates upward/offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the Earnest money deposited by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer/bidder for depositing the amount through MRO. Bids of such Contractors/bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Govt treasury. In addition, bids of such tenderer/bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tenderer/bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

5. **CPM (Critical Path Method):**

(a) The project planning for work covered in the scope of tender is based on CPM.

(b) The tenderer/bidder is expected to be fully conversant with the CPM technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literature on the subject are widely available in the market which the tenderer/bidder may make use of.

(c) The tenderer's /bidder's attention is drawn to special condition of the tender regarding preparation of the detailed network analysis and time schedule for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/bidder in using the technique will be taken as his technical inefficiency and will affect his class of enlistment and future prospect/invitation to tenders for future works.

6. Department may issue amendments/errata in form of CORRIGENDUM to tender/revised BOQ to the tender documents. The tenderer/bidder is requested to read the tender documents in conjunction with the errata/amendment/corrigendum, if any, issued by the department.

7. Bidders should check the website periodically for any modified/Revised BOQ uploaded by the Department through Corrigendum and it is bidder's responsibility to ensure uploading of correct BOQ. The bidder whose bid is not on correct BOQ shall be treated as NON-BONAFIDE.

8. These instructions shall form part of the contract documents.

**SIGNATURE OF CONTRACTOR**
**DATED: _____**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

**MILITARY ENGINEER SERVICES**
**NOTICE INVITING TENDER (NIT)**

1. A tenderunder e-procurement is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT) from the reputed and experienced contractors meeting the criteria as mentioned in Appendix 'A' to this Notice Inviting Tender (NIT).

2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work cost more or less a tenderer/bidder will have no claim on that account. The tender shall be based on as mentioned in aforesaid Appendix 'A'.

3. The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of Acceptance of tender.

4. Normally, contractors, whose names are on the MES approved list and within whose financial limit/class the estimated amount would fall, may tender/bid. Un-enlisted contractors, willing to participate in the bidding process, must meet the criteria specified in Appendix _A' to NIT.

5. In case, the tender submitted by enlisted as well as un-enlisted contractor is accepted, the contractor will be required to lodge **"Performance Security Deposit"**for an amount equivalent to **5% of the Contract Sum/Lump Sum** in favour of the Accepting Officer in the form of "**Bank Guarantee"** or **FDR.** Work Order No. 01 shall be placed only after submission of Performance Security of adequate value by the Contractor. In case, a Fixed Deposit Receipt (FDR) of any Bank is furnished by the contractor to the Government as part of the Performance Guarantee and the Bank is unable to make payment against the said Fixed Deposit Receipt, the loss caused thereby shall fall on the contractor and the contractor shall forthwith on demand furnish additional security to the Government to make good the deficit. The period of validity of the Bank Guarantee Bond against Performance Security shall be initially valid upto the stipulated date of expiry of Defects Liability Period plus minimum 60 days beyond that. In case final bill not paid during this period, the contractor shall get the validity of Performance Guarantee extended to cover such enlarged time required for payment of final bill.

6. Not more than one tender/bid shall be submitted/uploaded by one contractor or one firm of contractors. Under no circumstances will a father and his son(s) or other close relations, who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

7. The Office of **Commander Works Engineer (AF) Bagdogra** will be the Accepting Officer here in after referred to as such for purpose of the contract.

8. The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the tenderer/ bidder on or before the date & time mentioned in NIT. A scanned copy of DD with enlistment details/documents shall be uploaded as packet 1/cover-1 ('T bid) of the tender/bidone-tenderingportal.DDisrefundableincaseTbidisnotacceptedresultingin non-opening of Q bid. The applicant contractor shall bear the cost of bank charges for procuring and encasing the DD and shall not have any claim from Government whatsoever on this account.

**MILITARY ENGINEER SERVICES**
**NOTICE INVITING TENDER (NIT)**

8.1 Tender form and conditions of contract and other necessary documents shall be available on www.defproc.gov.inonlinesite for download and shall form part of contract agreement in case the tender/bid Is accepted.

8.2 In case of contractor who has not executed the Standing Security Bond, the Cover- I shall be accompanied with "Bid Securing Declaration" as mentioned in here-in-before at para 1, sub para (i) & (ii) of instructions on filling and submission of tender.

8.3 Copies of the drawings and other document pertaining to the work signed for the purpose of identification by the Accepting Officer or his accredited representative, sample of materials and stores to be supplied by the contractor will also be available for inspection by the tenderer/bidder at the office of Accepting Officer and concerned Garrison Engineer during working hours.

9. The tenderers/ bidders are advised to visit the site of work by making prior appointment with Garrison Engineer who is also the Executing Agency. The tenderers/bidders are deemed to have full knowledge of all relevant documents, samples, site etc., whether they have inspected them or not.

10. Any tender/bid which proposes any alteration to any of the conditions laid down or which proposes any other condition or prescription whatsoever, is liable to be rejected.

11. The uploading of bid implies that bidder has read this notice and the Conditions of Contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores, tools and plants etc. will be issued to him and local conditions and other factors having bearing on the execution of the work.

12. Tenderers/ bidders must be in possession of a copy of the MES Standard Schedule of Rates (see appendix 'A') including amendments and errata thereto.

13. Invitation for e- tender does not constitute any guarantee for validation of 'T' bid and subsequent opening of finance bid of any applicant/bidder, even of enlisted contractors of appropriate class, merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the 'T' bid and not open the finance bid of any applicant/ bidder. 'T' bid validation shall be decided by the Accepting Officer based on, inter alia, capability of the firm as per criteria given in Appx 'A' to this NIT. The applicant contractor/bidder will be informed regarding non-validation of his 'T' bid assigning reasons thereof through the www.defproc.gov.in website. **The applicant contractor/ bidder if he so desires may appeal to the next higher Engineer authority i.e. HQ Chief Engineer (AF) Shillong Zone one mail id: cezafs2-mes@nic.inwith copy to the Accepting Officer on email: cweafbagd3-mes@gov.in** before the scheduled date of opening of Finance Bid. The decision of the Next Higher Engineer Authority (NHEA) shall be final and binding. The contractor/ bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

**MILITARY ENGINEER SERVICES**
**NOTICE INVITING TENDER (NIT)**

14. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking, giving a price preference over other Tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible from such tenderer/ bidder whose tender/ bid is rejected.

15. The Court of the place from where tender has been issued shall alone have jurisdiction to decide any dispute out of or in respect of this tender. After acceptance of tender, Condition 72 (Jurisdictions of Courts) of IAFW- 2249 shall be applicable.

16. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

17. This Notice Inviting Tender (NIT) including Appendix 'A' shall form part of the Contract.

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

**(Signature of Contractor)**

**APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)**

| | | |
|---|---|---|
| 1. | Name of work | **PROVN OF WIDENING OF ROAD OF CERTAIN INTERNAL ROAD AND ALLIED WORKS AT AF STN BAGDOGRA UNDER GE (AF) BAGDOGRA.** |
| 2. | Estimated cost | Rs. **43.50** Lakhs ( At Par Market) |
| 3. | Period of completion | **180** days |
| 4. | Cost of tender documents | Rs. 500.00 in the form of DD/Banker's cheque from any Scheduled/Nationalized bank in favour of **GE (AF) BAGDOGRA** and payable at **BAGDOGRA.** (Scanned copy to be uploaded on-line and original to be forwarded off-line so as to reach this HQ before due date.) |
| 5. | Website/portal address | www.defproc.gov.in and www.mes.gov.in |
| 6. | Type of contract | The tender shall be based on IAFW-1779A(R) and IAFW-2249 where the rate is to be quoted by the contractor. |
| 7. | Timeline Details :- | **Refer critical dates** |
| | (a) Bid submission start date | |
| | (b) Bid submission in date | |
| | (c) Date of bid opening | |
| 8. | **Eligibility criteria** | |
| | (a) For MES enlisted contractors | Contractor shall be enlisted with MES in Class **'D'** and above and category **'a(v)' as per new Manual on Contracts 2020'** subject to satisfactory remarks wrt performance in respect of works in hand as reflected in Work Load return (WLR) or any other report circulated by competent engineer authority . |
| | (b) For other contractors | (i) Contractor not enlisted with MES should meet the enlistment criteria of **'D'** Class **'a(v)' as per new Manual on Contracts 2020'** category contractor with regard to satisfactorily completion of requisite value works with Central/State Government/ Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO / DGMAP, annual turnover, bank solvency, working capital and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website (www.mes.gov.in).<br>(ii) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES.<br>(iii) Not suspended /debarred /blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department or any Central/State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on the bid submission end date.<br>(iv) Details of works completed and under progress in MES be submitted in the following format:- |

| Srl No. | CA No. & Name of work | Value of CA | Date of commencement | Date of completion | Extended date of completion |
|---|---|---|---|---|---|

**APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (Contd…..)**

| | | |
|---|---|---|
| | | (v) Un-enlisted contractor who have secured **two works** in MES should **get themselves registered** in the appropriate designated Class with any Registering Authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with the MES. |
| 9. | Tender issuing and Accepting Officer | Name: HQ CWE (AF) BAGDOGRA<br>Address: PO- BAGDOGRA AIRPORT, Distt: Darjeeling (WB)<br>Pin:– 734 421<br>Phone No: 0353 2698308<br>Email Id: cweafbagd3-mes@gov.in |
| 10 | Executing agency | **GE (AF) BAGDOGRA** |
| 11 | Earnest Money | **Rs. 87,000.00** in favour of **GE (AF) BAGDOGRA** in the form of Deposit at call receipt, FDR not acceptable. (Scanned copy to be uploaded on-line and original to be forwarded off-line so as to reach this HQ before due date.) |

**NOTES** :

1. After opening of Cover 1, in case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as eligible un-enlisted contractors, if any, fulfilling the other criteria given in NIT is less than 7 (Seven), applications in respect of MES contractors of one class below the eligible class shall also be considered subject to fulfillment of other eligibility criteria given in the NIT. Therefore MES contractor's one class below may also bid for this tender. Such contractors (Contractors of one class below the eligible class) shall not be considered in case their present residual work in hand is more than FIVE TIMES their present tendering limit. However in case such contractors fulfill the criteria of up gradation to the stipulated eligible class based on past experience of completed works (individual work experience and/or average annual turnover, as applicable) and financial soundness (solvency/ financial soundness and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover-1 bid details related to residual work in hand like details of works in hand showing names of work, names of Accepting Officers, Contract amount, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfill the criteria of upgradation shall also upload the requisite information/ documents in support of upgradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7(Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of upgradation in respect of past experience of completed works (individual work experience and/or average annual turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore such contractors shall upload the requisite information/ documents in the Cover-1.

3. Unenlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign national/ Indian nationals staying abroad/ Indian national having taken foreign citizenship, as director (s) shall be considered subject to security clearance from the concerned authorities.

4. Contractors enlisted with MES will upload following documents in Cover 1 for checking eligibility:-

(a) Application for tender on Firm's letterhead.

(b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.

**APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (CONTD…)**

(c) Scanned copy of DD/ Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.

(d) Any other document required as described in this Appendix.

5. Contractors not enlisted with MES will be required to upload following documents in Cover 1 for checking eligibility:-

(a) Application for tender on Firm's letterhead.

(b) Scanned copy of DD/ Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.

(c) Copy of Police Verification Report/Police Clearance Certificate/ Character Certificate from the Police Authority of the area where the registered office of the firm is located/ notarized copy of valid passport of Proprietor/ each Partner/each Director.

(d) All documents required for enlistment in MES for the class mentioned in Para 8(b) above as per Para 1.5 of Section 1 of MES Manual on Contracts 2020.

(e) Details of works being executed in MES, if any.

(f) Any other documents required as described in this Appendix.

6. Tenders not accompanied by scanned copies of requisite DD/ Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover 1 shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.

7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting Officer within 05 days of bid submission end date failing which following action shall be taken.

(a) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non-submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial Bid (Cover 2).

(b) In case of tenders from unenlisted contractor, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of Financial bid (Cover 2).

(c) In case of tenders form enlisted and unenlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover 2).

8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/ another firm on his behalf. However a contractor can execute the work through power of attorney to sons/daughters/spouse of Proprietor/Partner/Director and firm's own employees, director, project manager **provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director**.

9. After opening of Cover 1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of e-mail/SMS/Sped Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same.

**APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (CONTD…)**

10. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA) i.e. HQ Chief Engineer (AF) Shillong Zone on email id cezafs2-mes@nic.in with copy to the Accepting Officer on email before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

11. In case an unenlisted contractor is already executing works in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES Class of Contractor for which it is eligible. For this purpose, details of the works being executed by such a contractor shall be uploaded in the Cover 1 of the bid and shall be checked/ verified by the Accepting Officer.

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (i.e. he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid/ bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

13. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a willful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an unenlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/ administrative action shall be taken against such tenderes. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any shall not be eligible for this tender in second call or subsequent calls.

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/ one or more Partners/ Directors are common. Decision of Accepting Officer on issue/ deny the tender to any one of the related firms shall be final and binding.

_____ _____

Signature of contractor DCWE (Contract)
For Accepting Officer

File No. 8549/HAS/2023-24/ /E8 Dated: 10 Jan 2024

HQ Commander Works Engineers (AF)
Bagdogra
Post : Bagdogra Airport
Distt : Darjeeling (WB)
PIN : 734 421

**In lieu of IAFW-1779A (Revised 1955) (To be used in conjunction with General Condition of Contracts, IAFW-2249, 1989 Print)**

**MILITARY ENGINEER SERVICES**

TELE: 03712-258155

HQ Commander Works Engineer
(AF) Bagdogra
Post : Bagdogra Airport
Distt : Darjeeling (WB)
Pin : 734 421

**8549/BAG/2023-24/____/E8** **26 Aug 2025**

**TENDER AND ITEM RATE CONTRACT FOR WORKS BY MEASUREMENT IN THE EXECUTION OF: PROVN OF WIDENING OF ROAD OF CERTAIN INTERNAL ROAD AND ALLIED WORKS AT AF STN BAGDOGRA UNDER GE (AF) BAGDOGRA.**

1. Shri/M/s_____ of _____ _____ is/are hereby authorized to Tender for the above work. The complete tender is to be uploaded in the www.defproc.gov.in site. The Cover-1 (Technical bid) and Cover-2 (Finance Bid) will be opened online only as per critical dates given.

2. Any correspondence concerning this tender should be addressed as indicated at the top of this sheet quoting the reference as given.

**THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER**

**SIGNATURE OF OFFICER ISSUING TENDER**
**APPOINTMENT : DCWE (CONTRACTS)**
**DATED : 26 Aug 2025**

**SIGNATURE OF THE CONTRACTOR**
**DATED_____**

**SCHEDULE `A' (BOQ) NOTES**
**(IAFW 1779-A REVISED 1955)**
**(LIST OF WORKS AND PRICES)**

**NAME OF WORK:PROVN OF WIDENING OF ROAD OF CERTAIN INTERNAL ROAD AND ALLIED WORKS AT AF STN BAGDOGRA UNDER GE (AF) BAGDOGRA.**

**(A) NOTES (GENERAL) APPLICABLE TO ALL SECTIONS:**

1. The quantities shown in column 3 are approximate and are inserted as a guide only. However, these shall not be varied beyond the limits laid down in condition 7 of IAFW-2249 General Condition of contracts.

2. Tenderer is to quote his unit rates under column 5.

3. The rates shall be quoted in figure only. Software system will automatically inserted base on the rate inserted by the bidder/ tenderer.

4. **SCHEDULE OF CREDIT**: Total amount under column 8 is not firm but will be treated as "CONTRACT SUM" referred to in IAFW-2249.Tenderer has to visit the BOQ pages of tender for quoting the rates of each items of the Schedules. Total amount of Schedule of credit has been catered towards Sch. of credit for which details have been made/ quoted by bidders in a separate schedule in the PDF document and tenderer shall study the same before quoting. The amount of schedule of credit shall be deducted from the total amount of BOQ under col-8. Contract sum shall be amended accordingly of lowest tenderer (L-1). Contractors are requested to follow note of this while quoting for item contained in BOQ (Cover-2). No cognizance on account of any variation shall be entertained at later date.

5. Total amount under column 8 is not firm but will be treated as "CONTRACT SUM" or all the purpose as referred to in IAFW-2249.

6 **UNIT RATE**: The tenderer shall calculate his own details and insert unit rates in Column 5 against each item of "Schedule of Works"/Schedule 'A' (BOQ). It is an express condition of this contract that the rates quoted by the tenderer in the tender for various items shall be deemed to include for the entire and final completion of the items of work in accordance with provision of the tender. Government shall not entertain any claim whatsoever on account of inaccuracies/misunderstanding, if any in the aforesaid rates.

7. Tenderers shall be deemed to include in his unit rate for each item for the provision of all materials, labour, operations, tools/plants as required and other requirements as detailed in the Particular Specifications irrespective of whether these appear as specific item or mentioned in the description of items in "Schedule of Works"/Schedule 'A' (BOQ) or not and the tenderer shall not be paid any extra for the same.

8. The under mentioned remarks shall be deemed to have been against each item of BOQ:-

   (a) Refer list of drawing
   (b) Refer note No : 11 of Schedule 'A' Notes
   (c) Refer notes to Schedule 'A.

9. **PERIOD OF COMPLETION**: Total work under this contract shall be completed within **180 Days** from the date of handing over of site.

10. Unless otherwise specified method of measurement of various items of work shall be as stipulated in MES standard Schedule of Rates (Part I – 2009 and Part II – 2020).

11. The rates quoted by the tenderer shall be deemed to include any minor details/items of work and/or construction which are obviously and fairly intended and which may not have been included in these documents but which are essential for the execution and entire completion of the work and decision of the Accepting officer on this account shall, be final conclusive and binding.

**SCHEDULE `A' (BOQ) NOTES (CONTD……./)**

12. **TAXES, DUTIES & LEVIES, ETC.**

12.1. The contractor's quoted rates shall be deemed to be include of all taxes/cesses viz GST, duties, royalties, octroi and other levies payable under respective statues as applicable on the date of receipt of tender. It may be noted that any tender imposing any condition in this regard or on any other account shall be treated as a conditional tender and the same shall be liable to be rejected. Also refer special condition in this regard.

12.2. No increase/decrease in any taxes/duties, statutory or otherwise after submission of tender and during the execution of the works shall be adjusted except as provided in special condition relevant clause herein after**.**

12.3 Cost of Testing including Transportation of Materials or Equipment's or Items, provision of all facilities for Testing in accordance with Specifications and/or IS shall be borne by the Contractor Unless otherwise mentioned.

12.4 **SAFETY MEASURES, PRECAUTIONS, RISKS, ETC.:-**

12.4.1 The work shall be carried out with almost care to ensure that no damage to Existing/ Adjoining work is done failing which the damage, if any done shall be rectified by the Contractor to match with the Existing/ Adjoining work to the entire satisfaction of Engineer-in-Charge under Contractor's own arrangement and at his own expenses.

12.4.2 Suitable Tools, Plants, Equipment's, Mechanism, etc., as considered necessary shall be adopted during execution of the work. The Contractor shall take all Precautions, Safety Measures, etc., to avoid a Damage, Miss-happening, Accident, etc., to the Workmen engaged by him to carry out the work. The UNIT RATES quoted by Contractor/ bidder shall be deemed to have included the element of adopting of safety measures, precautions and also the risks, etc. involved in work and nothing extra shall be admissible on this account.

13 **METHOD OF MEASUREMENTS**

13.1 The method of Measurements to be adopted in the Contract shall be as laid down in MES STANDARD SCHEDULE OF RATES PART-I & II, IS-1200 and other relevant BIS standard. The General Rules, Special Conditions, Preambles and Other Provisions in MES STANDARD SCHEDULE OF RATES PART-I & II and relevant BIS standard shall be deemed to apply to the Work under this Contract unless otherwise mentioned in the Tender documents. However, where the UNITS of Work given in Column 4 against Items of "Schedule of Works"/ Schedule 'A' (BOQ)are at variance with that mentioned against respective Items in MES STANDARD SCHEDULE OF RATES, the Measurements of work against such Items of "Schedule of Works"/ Schedule 'A'(BOQ)shall be done as per the UNITS given in "Schedule of Works"/ Schedule 'A'(BOQ).

14. Unless otherwise specifically mentioned, unit rate of each item of works is inclusive of all 'Materials and Labour' or 'Supply and Fixing' or 'Supply and Laying' including tools and plants, installation, commissioning and testing complete as required for entire completion of work.

15. The contractor's quoted rates shall be deemed to be inclusive of all taxes/cesses viz GST, duties, royalties, octroi and other levies payable under respective statues as applicable on the date of receipt of tender. It may be noted that any tender imposing any condition in this regard or on any other account shall be treated as a conditional tender and the same shall be liable to be rejected. Also refer special condition in this regard.

16. The work site lies at **Bagdogra** within the district of **Darjeeling** in the state of **West Bengal**. Tenderer's attention is invited to deduction of GST at source for execution of works contract as levied by the Govt. The rate quoted by the tenderer in Schedule 'A' shall be deemed to include for the above mentioned Tax and all other taxes levied by the Govt/Statutory body and nothing extra will be payable on this account.

**SCHEDULE `A' (BOQ) NOTES (CONTD……. /)**

17. The tenderer shall satisfy himself by visiting the site with prior appointment with GE/AGE (I) regarding working conditions, local conditions, nature of site/ground, climatic conditions, nature of work and means to access to site and other issues/circumstances which may affect the execution of work at a later date. No extra payment with regard to afore said reasons shall be admissible under any circumstances.

18. Retrieved materials (due to dismantling/demolition/ taking down etc) of serviceable nature shall be removed from site of work after deducting credit amount from RAR. Unserviceable retrieved materials and other debris shall be removed off from site to a distance exc. 50 mtr and not exc. 500 mtr and deposited as directed by Engineer-in-charge and the unit rate quoted for the dismantling/demolition/taking down etc. shall be deemed to be inclusive for above provision.

19. All the Code of Practice, Standards and Specifications applicable shall be the latest editions with up to date correction slip etc.

20. Please note that No foreign exchange or import license facility will be arranged by the department for any material included in the subject tender.

21. **Security/Restricted Area:-**The Works under this contract lies in **'RESTRICTED'** Area. The contractor, his agents, work people etc. shall adhere to this security instruction as laid down in the Contract Conditions strictly. No claim whatsoever for observance of restrictions imposed at site shall be admissible at any stage.

22. **INSPECTION OF THE WORKS (DEFECTS LIABILITY PERIOD)** (Refer Condition 46 of IAFW – 2249 (General Conditions of Contracts)):-
**The Defects liability period for the scope of work(s) covered under the contract is TWENTY FOUR CALENDAR months**. Any defects in the work noticed during this period shall be rectified by the contractor forthwith, on demand in writing from the GE specifying the area affected not withstanding that the work executed might have been inadvertently passed, certified and paid. The contractor shall carry out the rectification of the defects notified at his own expense during the defects liability period and in event of his failure to do within a period to be specified by the GE in his demand, the GE shall undertake such rectification work at the risk and expense of the contractor.

**SIGNATURE OF THE CONTRACTOR** **DCWE (CONTRACTS)**
**DATED:** **FOR ACCEPTING OFFICER**

## GENERAL SUMMARY

1. Total Amount of Schedule of works (BOQ) B/F from Serial Page No. 21 Rs._____

2. Total amount of Schedule of credit **Rs. _____(Minus)**

_____

**TOTAL CONTRACT SUM** **Rs.**

_____

(Rupees……..

……)

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

## SCHEDULE 'B'

## LIST OF MATERIALS TO BE ISSUED TO THE CONTRACTOR
## (REFER CONDITION 10 OF GENERAL CONDITIONS OF CONTRACTS - 1AFW-2249)

| Ser No | Particulars | Rate at which materials will be issued to the contractor | | Place of issue (By name) | Remarks |
|---|---|---|---|---|---|
| | | Unit | Rate | | |
| 1 | 2 | 3 | 4 | 5 | 6 |

**– NIL –**

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

## SCHEDULE 'C'

## ISSUE OF TOOLS AND PLANTS (OTHER THAN TRANSPORT) WHICH WILL BE HIRED BY THE CONTRACTOR
## (REFER CONDITION 15, 34 AND 35 OF IAFW-2249)

| Ser. No. | Quantity | Particulars | Details of MES Crew supplied | Hire charges per unit per working day | Stand by charges per unit per day | Place of Issue | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

**– NIL –**

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

## SCHEDULE 'D'

## TRANSPORT TO BE HIRED BY THE CONTRACTOR
## (REFER CONDITION 16, AND 35 OF IAFW-2249)

| Sr. No. | Quantity | Particulars | Rate per unit per working day | Stand by charges per unit per day | Place of Issue | Remarks |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

**– NIL –**

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

# TENDER

To

The President of India

Having examined and perused the following documents: -

1. Specifications signed by DCWE (CONTRACTS) for CWE (AF) Bagdogra.

2. Drawings details in the specifications.

3. Schedule 'A', 'B', 'C', and 'D' attached hereto.

4. MES Standard Schedule of Rates Part I, 2009 (specifications) & Part II (SSR -2020 Rates), (here-in-after and in IAFW-2249 (1989 Print), referred to as the MES Schedule) together with the following errata and amendments as under:-

(a) MES Standard Schedule of Rates Part I, 2009 (specifications) – 1 to 3

(b) MES Standard Schedule of Rates Part II, 2020 (Rates) - 1 to 122

5. General Conditions of contracts, IAFW-2249 (1989 print) together with errata No 1 to 20 and amendments 1 to 49.

6. Water will be supplied by the MES. Water supply shall be paid for by the contractor to the department @ **Rs. 3.75 per Rs. 1000/-** worth of work done.

Should this tender be accepted, I/We agree

(a) That the sum of **Rs. _____**forwarded as Earnest money, shall either be retained as a part of Security Deposit or be refunded by the Government on receipt of the appropriate amount of Security Deposit, all as per Condition 22 of IAFW-2249.

b) To execute all the works referred to in the said documents upon the terms and conditions, contained or referred to therein at the item rates contained in the aforesaid Schedule 'A' or at such other rates as may be fixed under the provision of condition 62 of IAFW-2249 and to carry out such deviations as may be ordered vide Condition 7 of IAFW-2249 up to a maximum of **±10% (TEN PERCENT)** and further agree to refer all disputes, as required by Condition 70 & 71 of IAFW-2249 to the sole arbitrator of a serving Officer having degree in Engineering or equivalent or having passed final/direct final Examination of Sub-division II of Institution of Surveyors (India) recognized by the Government of India to be appointed by the **ADG(NEI)** or in his absence the Officer Officiating as **ADG(NEI)** whose decision shall be final, conclusive and binding.

* To be deleted where NOT applicable.

Signature _____ Date _____

In the capacity of _____

duly authorized to sign the TENDER FOR AND ON BEHALF OF _____

_____)

(IN BLOCK CAPITAL LETTERS)

Name of the Signatory_____

(In Capital Letters)

Dated _____

POSTAL ADDRESS _____
_____
_____
Telephone No. _____
Mobile No. 1. _____Fax No._____
E-Mail ID 1. _____

Witness: _____
Address _____
_____
_____

**A C C E P T A N C E**

_____Alteration(s) have been made in these documents and as evidence that these alterations were made before the execution of the Contract Agreement and these have been initialed by the Contractor and_____**DCWE (CONTRACTS)**
.

The said officer is hereby authorized to sign and initial on my behalf the documents forming part of this contract.

The above tender was accepted by me on behalf of the President of India at the item rates contained in Schedule 'A' for Rs. _____ on _____ day of _____2025.

Signature _____Dated this _____ day of _____2025.

Appointment: COMMANDER WORKS ENGINEER (AF) BAGDOGRA
ACCEPTING OFFICER
(For and on behalf of the President of India).

**GENERAL CONDITIONS OF CONTRACTS (IAFW-2249: PRINT – 1989)**

**FOR**

**ITEM RATE CONTRACTS (IAFW-1779A)**

A copy of the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249-1989 print) has been supplied to me/us and is in my/our possession. I/We have read and understood the provisions contained in the aforesaid GENERAL CONDITIONS OF CONTRACTS before submission of this tender and I/We agree that I/We shall abide by the terms and conditions therefore, as modified, if any, elsewhere in this tender documents.

It is hereby further agreed and declared by me/us, that the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249-1989 Print) including condition 70 & 71 thereof pertaining to settlement of dispute by Arbitration, containing 33 pages (**Ser Page No 27 to 59**) with errata 1 to 20 and amendment Nos 1 to 49 (**Ser Page No 60 to 78**) form part of this tender documents.

It is also agreed by me/ us that in case of any discrepancy in the interpretation of the Contents between ENGLISH and HINDI VERSION, ENGLISH VERSION shall take precedence over HINDI VERSION.

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

Note: - In case the tenderer is not in possession of any of the documents referred above he should obtain a copy from the office of the HQ Commander Works Engineer (AF) Bagdogra, Post : Bagdogra Airport, Distt : Darjeeling (WB), Pin : 734 421, before submission of his tender.

**MILITARY ENGINEER SERVICES**
**ERRATA TO GENERAL CONDITIONS OF**
**CONTRACT IAFW - 2249 (1989 PRINT)**
**ERRATA**

| Errata No | Page No | Particulars |
|---|---|---|
| 1 | 2 | 3 |
| 1. | 4 | Condition 3, Para 3, Line 1: For “An’ Read “ and” |
| 2. | 7 | Condition 7, last Para, line 9 ; For “Objection” Read “objection” |
| 3. | 7 | Condition 7, last Para, line 11 : For “thetein” Read “therein” |
| 4. | 8 | Condition 8, line 7 “Delete” after the word “necessary” |
| 5. | 10 | Condition 15, Para 2, line 3, For “actory” Read “ Factory” |
| 6. | 11 | Condition 15, Para 2, line 6 For “escept” Read “except” |
| 7. | 11 | Condition 15, Para 2, line 6 For “authorise Read “ authorised” |
| 8. | 11 | Condition 15, Para 5, line 1, For “Maintence” Read “ Maintenance” |
| 9. | 16 | Condition 32, Para 1, line 4 For “sitning” Read “siting” |
| 10. | 16 | Condition 32, Para 2, line 6 For "reinstead" Read"reinstated" |
| 11. | 16 | Condition 34, line 6 For "revese " Read "revest" |
| 12. | 16.17 | Condition 36, para 2, line 2 For "conbusitible" Read"combustible" |
| 13. | 17 | Condition 37, line 3 For "whese " Read "whose" |
| 14 | 17 | Condition 43,line 1 For "shal" Read " Shall" |
| 15. | 17 | Condition 44, para 2, line 4 For"visit this " Read "visit his" |
| 16. | 18 | Condition 48, para (a), line 5 For "airising" Read "arising" |
| 17 | 20 | Condition 54, line 1 and 2 illigible word is "Contractor" |
| 18. | 20 | Condition 54, para below (a), Insert`b' in the exsiting bracket "( )" |
| 19. | 30 | ANNEXURE `B', Clause 6, line 2 Delete the word "after" |
| 20. | 30 | ANNEXURE `B', Clause 6 but line 3 Insert fullstop"(.)" after the word "thereof" and For "laddor" Read "Ladder" |

**SIGNATURE OF THE CONTRACTOR** **DCWE (CONTRACTS)**
**DATED:** **FOR ACCEPTING OFFICER**

## MILITARY ENGINEER SERVICES
## AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)

| Amendment No | Page No | Particulars |
|---|---|---|
| 1 | 2 | 3 |
| 1. | 28 & 29 | Annexure `A' Military Engineer Services Contractor's labour Regulations<br><br>(a) Para 2 (a) line 3<br>For : Not exceeding Rs.400 per month<br>Read: Not exceeding Rs.500 per month"<br><br>(b) Para 11, line 5 Add the following after the words labourers concerned. "The Garrison Engineer shall payments to the labour concerned within 45 days from the receipt of the report from the labour welfare officer or other person authorised as aforesaid, as the case may be." |
| 2 | 9 | Condition 10(R), Line 2 to 6.<br>For: Materials which_____ as aforesaid.<br>Read: "Materials which Govt shall supply are shown in Schedule `B' which also stipulates place of issue and rate(s) to be charged in respect thereof". |
| 3. | 19 | Condition 50, Sub para 4(b) (iii) Delete the existing description against sub para 4(b) (iii) and insert as under:-<br><br>(iii) Where the contract sum exceeds Rs.20lakhs but does not exceed Rs. 150 lakhs. — Seven and half percent of the Contract value of the item or group of items of work for which a separate period of completion is given or Rupees seven and a half lakhs whichever is less.<br><br>(iv) Where contract exceeds Rs.150 lakhs but does not exceed Rs. 300 Lakhs. — Five percent of the contract value of the item or group of the items of work for which a separate period of completion is given or Rupees twelve and a half lakhs whichever is less.<br><br>(v) Where the contract sum exceeds Rs. 300 lakhs. — Four percent of the contract value of the item or group of items of work for which a separate period of completion is given subject to a maximum of Rupees twenty five lakhs." |
| 4. | 27 | CONDITION 70 TO SUB PARA-9<br>Add the following at the end of the sub para : "If the value of the claims or counter claims in an arbitration reference exceeds Rs. 1 Lakh the arbitrator shall give reasons for the award." |
| 5. | 14 &15 | CONDITION 25<br>Delete para 2 and 3 of the above condition and substitute the following :- "Where the contractor is not a qualified Engineer or even if he is so qualified, he cannot, in the opinion of the GE give his full personal attention to the works, he shall at his own expense employ a person/person(s) possessing the following qualifications, and/or experience as his accredited agent(s)to supervise the works and to receive instructions from the Engineer-in-Charge:- |

**MILITARY ENGINEER SERVICES**

**AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)**

| Amendment No | Page No | Particulars |
|---|---|---|
| 1 | 2 | 3 |
| 5.(Contd..) | | (a) For works costing over Rs. 300 lakhs other than those covered in(g) below. — Two Degree holders in Engineering from Government recognised Institution or equivalent with at least 5 years practical experience of works.<br>and<br>Adequate numbers of Diploma holders in Engineering from a government recognized Institution with at least 8 years practical experience of works. |
| | | (Note: - The above provision shall be applicable irrespective of the fact whether contractor himself is a qualified Engineer or not). |
| | | (b) For works costing between Rs.100 lakhs and Rs.300 lakhs. other than those covered in (g) below — A degree holder in Engineering from a Government recognised Institution or equivalent with at least 4 years practical experience of works<br>OR<br>Two diploma holders in Engineering from a Government recognised Institution with at least 6 years practical experience of works |
| | | (c) For works costing between Rs.60 lakhs and Rs.100 lakhs other than those covered in (g) below — A degree holder in Engineering from a Government recognised Institution or equivalent with at least 3 years practical experience of works.<br>OR<br>A diploma holder in Engineering from a Government recognised Institution with at least 5 years practical experience of works. |
| | | (d) For works costing between Rs.7.5 lakhs and Rs.60 lakhs other than those covered in (g) below — A degree holder in Engineering from a Government recognised Institution or equivalent with at least 2 years practical experience of works.<br>OR<br>A diploma holder in Engineering from a Government recognised Institution with at least 4 years practical experience of works. |
| | | (e) For works costing between Rs.2 lakhs and Rs.7.5 lakhs other than those covered in (g) below — A degree or diploma holder in Engineering from a Government recognised Institution with adequate practical experience of works. |

**MILITARY ENGINEER SERVICES**

**AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)**

| 1 | 2 | 3 |
|---|---|---|
| | | (f) For works costing upto Rs. 2 lakhs other than those covered in (g) below. — competent person with adequate practical experience of works. |
| | | (g) For works of any value for repairs to buildings and roads, periodical Services to building, term contract, grass cutting and fencing — A competent person with adequate practical experience of works. The employment of Agent(s) as aforesaid shall be to the approval of GE who may verify his / their qualifications and experience by referring to original degree/diploma/testimonials which shall be made available to him by the contractor or by the individual employed or proposed to be employed. |
| **6** | 27 | Delete the description added at the end of sub para 9 vide amendment No.4 (1989 Print) and insert as under :-<br>"The arbitrator shall give reasons for the award in each and every case irrespective of the value of claims or counter-claims". |
| **7** | 24 | CONDITION 64<br>Delete para 4 viz `Provided the amount_____ shall not exceed Rs. 1.5 lakhs in toto. |
| **8.** | 26 | CONDITION 70 ARBITRATION PARA 1, LINE 4<br>Amend the words "Engineer Officer" to Read" Serving Officer having degree in Engineering or equivalent or having passed final/direct final Examination of of Sub-Division II of Institution of Surveyor (India) recognised by the Govt. Of India." |
| **9.** | 2 | Add the following new condition under chapter IV:-<br>"71. Jurisdiction of Courts". |
| | 27 | Add the following new condition:-<br>"71. Jurisdiction of Courts-Irrespective of the place of issue of tenders, the place of acceptance of tenders, the place of execution of contract or the place of payment under the contract, the contract shall be deemed to have been made at the place from where the acceptance of tenders has been issued and the work is executed/executable, The courts of the place from where the acceptance of the tender has been issued and the work is executed/under execution shall alone have jurisdiction to decide any dispute arising out of or in respect of the contract." |
| **10.** | 14 & 15 | CONDITION 25 AS AMENDED VIDE AMENDMENT No.5<br>FOR" Degree holders in Engineering from a Government recognised Institution or equivalent" wherever occuring.<br>READ "Degree holders in Engineering from a Government recognised Institution or equivalent/final or direct final pass of sub division II of the Institution of Surveyors (India)." |
| **11.** | 27 | CONDITION 70 : ARBITRATION PARA 8<br>FOR : "The Arbitrator may, from time..... Publishing the award."<br>READ: "The arbitrator may, from time to time with the consent of the parties, enlarge the time for making and publishing the award." |

**MILITARY ENGINEER SERVICES**
**AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)**

| 1 | 2 | 3 |
|---|---|---|
| **12.** | 24 | Add the following Para 4 after Para 3 (c) :-<br>"The amount so retained from the contractor shall be called retention money and shall be released to the contractor along with the final bill. However, in case the amount of this retention money is more than 1.50 lakhs, then after retaining an amount of Rs. 1.50 lakhs or 1% of the contract amount as executed whichever is more, the balance amount of retention money will be refunded to the contractor at satisfactory completion of works as certified by the GE." |
| **13.** | 23 | Condition 62 (G) valuation of deviation Para 1 line 3<br>For : "Plus 10 %" Read : "Plus 15 %" |
| **14.** | 5 | (a) Condition 4A(a), Line 4<br>FOR: "Rs. 2,500/-"<br>READ :"Rs.5,000/-"<br>(b) Condition 4A(d), Line 2 &3<br>FOR : "Rs. 5,000/-"<br>READ: 10,000/- |
| **15.** | 13 | Condition 22, Security Deposit, sub para 3, Line 3 and 4<br>For : "Controller of Defence Accounts concerned"<br>Read : "Accepting Officer" |
| **16.** | 24 | Condition 64, Sub para 7, lines 4 & 5<br>For:"Controller of Defence Accounts"Read:"Accepting Officer" |
| 17. | 19 | (a) Condition 50, Sub Para 2, Line 2a<br>For : " Deviation orders on the contractor"<br>Read : "Deviation order, escalation amount"<br>(b) Condition 50, Sub Para 4 (b) (i) to 4 (b) (v)<br>Delete existing sub para 4 (b) (i) to 4 (b) (v) in toto and insert as under ;<br>"Ten percent of the contract value of the item or group of items of work for which a separate period of Completion is given."<br><br>(c) Condition 50, Sub Para para 5<br>Add Sub para 5 as under<br>"Amount of retention money plus compensation during currency of contract shall not exceed ten percent of contract value. Contractor shall furnish BGB or Fixed Deposit Receipt in lieu of retention money and if compensation amount is not fully met from retention money, the difference if any shall be recovered from the payments due to the contractor." |
| 18. | 3 and 4 | (a) Condition 1(a), line 4<br>For : " the M.E.S. Schedule, the Specifications"<br>Read : "the M.E.S. Schedule, the Special Conditions, the Specifications"<br>(b) Condition 1(b), line 2<br>For : " these conditions, and the Specifications"<br>Read : "these conditions, the Special Conditions and the Specifications"<br>(c) Condition 1(b), line 3<br>For : " loaned"<br>Read : "sold" |

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| **18(contd..)** | | (d) Condition 1(g), line 3<br>For : " Garrison Engineer" who administers"<br>Read : "Garrison Engineer" (or Assistant Garrison Engineer (Independent) if applicable) who administers"<br><br>(e) Condition 1(g), line 3<br>In the end, add the following:-<br>" Where the office of the CWE does not exist and the GE is directly under the Chief Engineer, the authority of the CWE stipulated in these conditions shall vest in the concerned Chief Engineer."<br><br>(f) Condition 1(h), line 3<br>For : "the Superintendent Grade I"<br>Read: " the Junior Engineer (JE)"<br><br>(g) Condition 1(i), line 1<br>For : "Indian Standards Instiution."<br>Read: " Bureau of Indian Standards."<br><br>(h) Condition 1(p), line 4<br>For : "air craft and acts of God"<br>Read: "air craft and natural calamities"<br><br>(j) Condition 1(p), line 5<br>For : "floods and tornado."<br>Read: "floods, tornado and Tsunami." |
| **19.** | 5 | (a) Condition 4A, line 7<br>For : "Rs. 60,000/-"<br>Read: "Rs. 1,50,000/-"<br><br>(b) Condition 4A(a), line 4<br>For : "Rs. 2,500/-"<br>Read: "Rs. 10,000/-"<br><br>(c) Condition 4A(c), line 4<br>For : "Rs. 2,500/-"<br>Read: "Rs. 10,000/-"<br><br>(d) Condition 4A(d), lines 2 and 3<br>For : "Rs. 5,000/-"<br>Read: "Rs. 20,000/-"<br><br>(e) Condition 4A(f), line 1<br>For : "Rs. 60,000/-"<br>Read: "Rs. 1,50,000/-" |

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| 20. | 5 | (a) Condition 6, subpara 1<br>The existing contents shall be substituted as under:-<br>"**6. Provisional Items**- The amount pertaining to provisional items need not be deducted from the Contract Sum. The Engineer-in-Charge and Contractor shall set out the works covered under provisional items and provisional lump sum based on the description of items, drawings forming part of contract agreement and considering the ground conditions as encountered at site of works. The statement of variation in quantities, new items or deletion of items from the contract agreement as required will be worked out for approval of Engineer-in-Charge who will convey approval of such changes through site order book. For the purpose of payment the variation in value of work executed under these items shall be ascertained by measurement or valuation as for deviation. The variations shall be regularized as per condition 7." |
| 21. | 6 and 7 | (a) Condition 6A, line 6<br>Add following after the words 'preference to scale.'-<br>"However the provisions of any drawing of a later date shall take precedence over that of a drawing of previous date further subject to the condition that working drawing will take precedence over a Typical Detail Drawing."<br>(b) Condition 6 A(a), line 1<br>For: "Quantities."<br>Read: "Quantities, Schedule 'B', 'C', 'D',"<br>(c) Condition 6 A(a), subpara 3, line 1<br>For: "one document forming"<br>Read: "one document as defined hereinabove forming"<br>(d) Condition 6 A(A)<br>(i) Existing subpara (b) to (f) shall be enumerated as subpara (c) to (g)<br>(ii) Add new subpara (b) after existing subpara (a) as under:-<br>"(b). Items and their quantities for which no rates or prices have been quoted shall be deemed to have been covered by the rates and prices quoted for the other items of Schedule 'A' or bill of quantities."<br>(e) Condition 6 A(C) (iii)<br>The existing contents shall be substituted as under:- |

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| (iii) | Contract sum exceeding Rs. 10 lakh but not exceeding Rs. 100 lakh | : Rs. 20,000 plus 2% of Contract Sum exceeding Rs. 10 lakh subject to a maximum of Rs. 1,00,000/- |
| (iv) | Contract sum exceeding Rs. 100 lakh | : 1% of Contract Sum subject to a maximum of Rs. 5 lakh |

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| 22. | 8 | (a) Condition 9 (c), line 19<br>For: "labour at Site"<br>Read: "labour and tools and plants at Site"<br>(b) Condition 9(c), line 19<br>Add following at the end of para (c) viz line 21:-<br>"The contractor shall give the details of idle labour, workmen, employees, tools and plants and such other details as asked by the GE, within seven days of issue of order of suspension and the GE shall verify the necessity and correctness thereof. Proper record of these details shall be maintained duly signed by the GE and the contractor." |

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| 23. | 8, 9 and 10 | (a) Condition 10 A, subpara 2<br>Add in the end after the word "comply" following:-<br>"However if the cost of particular item of material in a contract exceed Rs. 1 lakh, these materials shall be procured only from the manufactures or from their authorised dealers/ stockist (except in case of material of local origin) and the contractor shall furnish proof thereof to the satisfaction of the GE that the material so comply."<br><br>(b) Condition 10 A, subpara 2<br>Add in the end after the word "in the Contract." following:-<br>"The approved samples of materials which loose their identity after incorporation in the work shall be preserved with the GE till completion of work. Thereafter the same shall be removed in 'as is where is' condition by the contractor with prior permission of the GE without any extra cost to the Government. However, the approved samples of material which do not loose their identity after incorporation and which can be incorporated in the work as decided by the Engineer-in-Charge shall be allowed to be used in the sample quarter/ block/ work."<br><br>(c) Condition 10 B, line 1 of last subpara of page 9<br>For: "cost of loading,"<br>Read: "cost of carriage, loading,"<br><br>(d) Condition 10 B, line 3 of subpara 3 of page 10<br>For: "the Engineer-in-Charge may,"<br>Read: "the GE may,"<br><br>(e) Condition 10 B, line 5 of subpara 3 of page 10<br>For: "as decided by the GE."<br>Read: "on the date of notifying to the contractor by GE (after technical check of final bill in CWE's Office) as decided by the GE or the stock book rate as fixed by the department, whichever is higher."<br><br>(f) Condition 10 B, line 6 of subpara 3 of page 10<br>For: "the market rate,"<br>Read: "the recovery rate," |
| 24. | 10 and 11 | (a) Condition 11(A), subpara (i) to (vii)<br>Existing contents of subpara (i) to (vii) shall be replaced as under with sub para (i) to (iv):-<br>"(i) by reason of civil commotion, local combination of workmen, strike or lockout, affecting any of the trades employed on the work, or<br>(ii) by reason of delay on part of nominated sub contractors, or nominated suppliers which the Contractor has , in the opinion of G.E., taken all practicable steps to avoid, or reduce, or<br>(iii) by reason of delay on the part of Contractors or tradesmen engaged by Government in executing works not forming part of the contract, or<br>(iv) by reason of any other cause (except force majeure) which in the absolute discretion of the Accepting Officer is beyond the Contractor's control;" |

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| 24 (Contd.) | | (b) Condition 11(A), last subpara, line 2<br>For: "G.E."<br>Read: "Accepting Officer"<br>(c) Condition 11(B), last but 3rd line from end<br>For: "G.E."<br>Read: "Accepting Officer"<br>(d) Condition 11(C)<br>The existing contents shall be substituted as under:-<br>"(C) Extension of time if due shall be granted within 45 days of receipt of request from the contractor along with supporting documents, but before expiry of original/ extended period of completion.<br>(D) No claim in respect of compensation or otherwise, for idle labour and/ or idle machinery etc. and/ or business loss or any such loss, howsoever arising, as a result of extensions granted under Conditions (A) and (B) above shall be admissible. The decision on reason and quantum of extension shall be final binding.<br>(E) **DELAY ON ACCOUNT OF FORCE MAJEURE –**<br>Should any force majeure circumstances arise, each of the contracting party will be excused for the non fulfilment or for the delayed fulfilment of any of its contractual obligations, if the affected party within 15 days of its occurrence informs the other party in writing.<br><br>Force majeure shall mean fires, floods, natural calamities such as earth quakes, lightening or other acts such as war, turmoils, strikes (otherwise than contractor's employees), invasion, act of foreign enemies, hostilities, civil war, rebellion, revolution, insurrection, military or usurped power, damage from aircraft, sabotage, explosions, quarantine restrictions, beyond the control of either party.<br><br>It is understood and agreed between the parties here to that the rights and obligations of the parties shall be deemed to be in suspension during the continuance of the force majeure even as aforesaid and the said rights and obligation shall automatically revive upon cessation of the intervening force majeure event. The period within which the rights and obligations of the parties shall be in suspension due to force majeure event, shall not be considered as a delay with respect of the period of completion and/ or taking over work under the contract or otherwise to the detriment of either party.<br><br>Not withstanding the provision of the immediately foregoing clauses, it is further understood and agreed between the parties hereto that in the event of any force majeure persisting for an uninterrupted period exceeding 6 (Six) months, either party hereto reserves the right to terminate this contract upon giving prior written notice of 30 (thirty) days to the other party of the intension to terminate without any liability other than agreement for the completed work and/ or contractor's materials lying at site." |
| 25. | 12 | (a) Condition 17, line 1<br>For: "The Contractor shall not"<br>Read: "The Contractor including Public Sector undertaking/ Government agency shall not" |

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| 26 | 13 | (a) Condition 18, line 1<br>For: "The Contractor shall not"<br>Read: "The Contractor including Public Sector undertaking/ Government agency shall not" |
| 27. | 14 and 15 | (a) Condition 25, the contents of subpara 2 after line 5 shall be substituted as under:- |

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| (a) | For works costing between Rs. 50 lakh to 1000 lakh | A Degree holder in Engineering from a Govt recognized Institution or equivalent, with final or direct final passed of Sub Division II of the Institution of Surveyors (India) with at least 4 years practical experience of works. |
| (b) | For works costing between Rs. 7.5 lakh to 50 lakh | A Degree holder in Engineering from a Govt recognized Institution or equivalent, with final or direct final passed of Sub Division II of the Institution of Surveyors (India) with at least 2 years practical experience of works. |
| (c) | For works costing below Rs. 7.5 lakh | A Diploma holder in Engineering from a Govt recognized Institution with adequate practical experience of works. |

Notes:-

(1) Engineers (Degree/ Diploma holders) employed should be of the relevant discipline to which nature of work pertains.
(2) The provision at Serial (b) & (c) above shall be applicable irrespective of the fact whether contractor himself is a Qualified Engineer or not.
(3) Contractor shall employ additional Engineers as directed by GE where there are scattered sites.
(4) For works costing more than Rs. 10 crore, the requirement of Engineering staff shall be as given in the tender documents.
(5) For specialist works/ services Accepting Officer may vary requirement of supervisory staff in tender documents.

(b) Condition 25, last subpara on page 15:
The contents of last subpara on page 15 shall be substituted as under:-

" The G.E. shall have full powers, to put the contractor on notice on account of default either for non-employment of Engineer(s) or absence of Engineer(s) from site and levy penalty @ Rs. 500/- per day per vacancy upto 30 days period. Thereafter GE shall have the option to either suspend the work or employ Engineer(s) at contractor's cost and recover the amount from contractor's dues."

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| 28. | 15 | (a) Condition 26, last subpara, last line<br>For: "Rs. 50/-"<br>Read: "Rs. 5,000/- (Rupees five thousand only)" |
| 29. | 16 | (a) Condition 31, third subpara, line 2<br>For: "at the *All India Flat Rate per 1,000 gallons which"<br>Read: "at the *All in cost Rate per 1,000 gallons subject to a minimum of Rs. 3.75 per every Rs. 1000/- worth of work done priced at contract rates which" |

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| 30 | 17 | (a) Condition 36, last subpara<br>Add the following at the end:-<br>" Fencing be provided wherever necessary as decided by GE to isolate the working area to make the area unrestricted from restricted" |
| 31. | 17 | (a) Condition 44, subpara 3, line 1<br>For: "rupees one lakh"<br>Read: "rupees two lakh"<br>(b) Condition 44, subpara 3, line 2<br>For: "rupees five lakh"<br>Read: "rupees ten lakh" |
| 32. | 18 | (a) Condition 46, subpara 2, line 11 to 14<br>For: "Provided always that the liability of the Contractor under this Condition shall not extend beyond the defects liability period except as regards workmanship which the G.E. shall have previously given notice to the contractor to rectify."<br>Read: "Alternatively, such work, if technically/ structurally acceptable, without detriment to the safety and utility of the item and the structure may be permitted to be accepted as devalued and recovery shall decided by competent authority (CWE in respect of contract concluded by himself and GE's and AGE(I) and CE in respect of contract concluded by him) or he may reject the work outright without any payment and/ or get it and other connected and incidental items rectified, or removed and re-executed at the risk and cost of the contractor. Whether any particular defect is due to unsound, imperfect or unskillful workmanship or due to normal wear & tear or user's negligence, decision of GE shall be final and binding. Provided always that the liability of the contractor under this Condition shall not extend beyond the defects liability period except as regards workmanship which the G.E. shall have previously given notice to the contractor to rectify. Govt. further reserves the right to get the work technically inspected during currency of the contract and also during defects liability period by the Additional Director General of Technical Examination and/ or his Officers or any other agency. The defects observed as a result of such technical examination shall be rectified by the contractor as notified by the GE. However, if the defects are not rectified, the devaluation of the work shall be carried out and recovery thereon shall be affected. " |
| 33. | 19 | (a) Condition 49, subpara 2, line 2,3 and 4<br>For: "within such period as may be notified by the Engineer-in-Charge, to the place of issue against written receipt from the Engineer-in-Charge."<br>Read: "to the place of issue as stipulated in Condition 10(B) here-in-before against written receipts from the Engineer-in-Charge."<br>(b) Condition 49, subpara 5, line 3,4 and 5<br>For: "before the completion of entire group, but for all purposes of the contract except for compensation for delay, the completion of the entire group shall be taken into account."<br>Read: "before the completion of the entire group. In such event, the grouping (phasing) of items as catered for in the contract shall be deemed to have been amended accordingly." |

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| 33. (Contd.) | | (c) Condition 49, subpara 7<br>Existing contents shall be substituted as under:-<br><br>"On receipt of notice from the Contractor that the work has been completed, the G.E. shall within seven days certify to the Contractor the Date(s) on which the items or group of items of works are completed and taken over and the state thereof or shall notify the details of incomplete items of work to the contractor. In case of dispute between G.E. and the Contractor over completion of work, the decision of Accepting Officer or CWE in case of G.E.'s contract shall be final and binding." |
| 34. | 21 | (a) Condition 55<br>Existing contents shall be substituted as under:-<br><br>"**55. Termination of Contract for Death**—Without prejudice to any of the rights or remedies under this contract, if the Contractor dies, the Accepting Officer shall have the option of terminating the Contract without compensation to the Contractor. If proprietor has nominated a person during his lifetime, the nominee will be allowed by the GE to complete the balance work. However if the nominee is not willing or in the opinion of Accepting Officer is not capable of completing the work as contracted for, he shall terminate the contract without any compensation to the nominee. The decision of the Accepting Officer whether the nominee is capable or not shall be final and binding." |
| 35. | 21 and 22 | (a) On page 2 against Sl 59, Blank, add following in description<br>"Determination of contract in the event of Force Majeure_"<br><br>(b) On page 22, against SL 59, Blank<br>Insert new condition 59 in lieu of **Blank** as under:-<br><br>"**59. Determination of Contract in the event of Force Majeure—** The contract may be determined at the option of either party by giving 30 days notice in writing to the other party should any event of 'Force Majeure' continue to prevail for an uninterrupted period of six months and no progress of work is achieved owing to such circumstances during these six months. On receipt of notice from one party to other explaining circumstances of 'Force majeure' the Garrison Engineer and contractor will carry out joint inspection of works and an inventory of completed, incomplete works and the materials collected for incorporation in the work shall be prepared for processing of final bill. The contractor shall have no claim to payment of any compensation on account of any profit or advantage which he may have derived from the execution of work in full but which he could not derive due to determination of contract on account of 'Force Majeure." |
| 36. | 22 | (a) Condition 61, Last but one para, line 6<br>For: "Rs. 500/-"<br>Read: "Rs. 5,000/-" |

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| 37. | 24 | (a) Condition 64, subpara1 and 2<br>Substitute existing contents with following:-<br><br>"**64. Advances on Account**-- The contractor may at intervals of not less than 30 days for contracts of value less than Rs. 50 lakhs and 15 days in case of contracts more than 50 lakhs submit claims on I.A.F.W. – 2263 provided the payment due is not less than Rs. 50,000/- and for Term Contracts, contractor may prefer not more than two claims for payment of advances on account of work done and of materials delivered in connection with Measurement and Lump Sum Contracts. However such claim for work done, which are required to be measured, shall be submitted only after recording joint measurements in the MES Measurement Book IAFW-2261."<br><br>(b) Condition 64, subpara 3, line 3<br><br>For: "Engineer-in-Charge:-"<br>Read: "Garrison Engineer:-"<br><br>(c) Condition 64, subpara 8 and 9<br>Insert a new subpara between subpara 8 and 9 as follows:-<br><br>"Provided further, the contractor may be paid advance on account to the full value of materials such as fittings and fixtures and other manufactured items as decided by the GE which do not lose their identity, brought on the site, on his furnishing Guarantee Bond(s) or Fixed Deposit Receipt(s) from Schedule Bank for the amount of retention money which should otherwise be recoverable from him under the contract. The Guarantee Bond and Fixed Deposit Receipt shall be executed and kept valid in a manner as described here-in-before." |
| 38. | 25 | (a) Condition 66, line 2<br><br>For: "the period being"<br>Read: "the period to be"<br><br>(b) Condition 66, line 3<br>For: "by the G.E."<br>Read: "by the Engineer-in-Charge"<br><br>(c) Condition 66(a)<br>For: "Four months"<br>Read: "Six months"<br><br>(d) Condition 66(b)<br><br>For: "Six months"<br>Read: "Nine months" |
| 39. | 27 | Existing Condition 71 Jurisdiction of Courts shall be renumbered as Condition 72 |

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| 40. | 27 | Add a new Condition 71 Conciliator as under:-<br><br>"**71. Conciliator**—If a dispute (other than those for which the decision of the CWE or any person is by the contract expressed to be final and binding) of any kind whatsoever arises between the parties to the contract during the execution of the works, or after completion or after determination cancellation/ termination of the contract, including any disagreement by either party with any action, inaction, opinion, instruction, certificate or valuation by the Accepting Officer or his nominee, the matter in dispute shall, in the first place be referred to the Disputes Resolution Board (DRB) in case of contracts valuing Rs. 10 crore or more and to conciliation, by a sole conciliator, in case of contracts valuing less than Rs. 10 crore. In case of disagreement with the decision of such DRB or conciliator, any party may invoke arbitration clause.<br><br>Procedure for the constitution/ appointment of DRB/ Conciliator shall be as laid down in the Contract Agreement." |
| 41 | 13 | **Condition No 22**, Security Deposit including all its sub para shall be deleted. |
| 42 | 20 | **Condition No 53**, Last Three Paragraphs after Para 53(c) : Existing contents shall be substituted as under:-<br>Whenever the Accepting Officer exercised his authority to cancel the contract under this condition, he may complete the Works by any means independently without risk and cost of the original contractor.<br><br>On cancellation of contract, the performance security and retention money upto last paid RAR shall be forfeited. All T&P and material of contractor lying at site shall be confiscated by Government and shall be absolutely at the disposal of the President of India and no compensation whatsoever shall be allowed by department.<br><br>If the failed contractor is a Company, or a Firm then every member / partner / Director of company or Firm shall be barred from participating in the tender for the balance work either in his / her individual capacity or as a partner of any other Company / Firm. |
| 43 | 20 & 21 | **Condition No 54**, Last Three Paragraphs after Para 54(d) : Existing contents shall be substituted as under:-<br>The Accepting Officer may without prejudice to any other right or remedy which shall have accrued or shall accrue thereafter to the Government, cancel contract as a whole or only such work order (s) or items of work in default from the contract. Whenever the Accepting Officer exercises his authority to cancel the contract under this condition, he may complete the works by any means independently without risk and cost of the original contractor.<br><br>On cancellation of contract, the performance security and retention money upto the last paid RAR shall be forfeited. All T&P and material of contractor lying at site shall be confiscated by Government and shall be absolutely at the disposal of the President of India and no compensation whatsoever shall be allowed by department. |

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| | | If the failed contractor is a Company, or a Firm then the every member / partner / Director of Company, or Firm shall be barred from participating in the tender for the balance work either in his / her individual capacity or as a partner of any other Company / Firm. |
| 44 | 25 & 26 | (i) **Condition No 67,Para 67(a), Line - 10** :-<br>**For**:-"Security Deposit or Security Bond amount"<br>**Read**:-"Performance Security amount (or from the Contractor's Standing Security Deposit amount or Security Bond if Performance Security amount is not adequate."<br><br>(ii) **Para 67(g), Line - 5:-**<br>**For** :-"Security Deposit or Security Bond amount"<br>**Read**:-"Performance Security amount (or from the Contractor's Standing Security Deposit amount or Security Bond if Performance Security amount is not adequate." |
| 45 | 26 | **Condition No 68,** Refund of Security Deposit including all sub paras : Existing contents shall be substituted as under:-<br><br>**Refund of Performance Security:** The Performance Security Deposit mentioned in Condition 19 above may be refunded to the contractor after the expiration of the defects liability period (vide Condition 46) by the GE provided always that the contractor shall first have been paid the final bill and have rendered a No-Demand Certificate (I.A.F.W - 451) |
| 46 | 26 | **Condition No 70,** Para 3:-<br>Delete Para 3 viz"Provide that in the ……. through any other contractor or Contractors or Agency or Agencies." |
| 47 | 1 | On page No - 1, Condition No - 19, Add "Performance Security" in lieu of "Blank" |
| | 13 | 19. **Performance Security**<br>19.1 Within 28 days of receipt of the Letter of Acceptance, the successful contractor shall deliver to the Accepting Officer a Performance Security in any of the forms given below for an amount equivalent to 5% of the contract sum.<br><br>(a) A Bank Guarantee in the prescribed form.<br><br>(b) Government Securities, FDR or any other Government Instruments stipulated by the Accepting Officer.<br>19.2 If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, it shall be issued by Nationalized / Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.<br><br>19.3 Failure of the successful contractor to comply with the requirements of sub-clause 19.1 shall constitute sufficient grounds for cancellation of the award of work and forfeiture of the Earnest Money, in case of MES enlisted contractor, amount equal to the Earnest Money stipulated in the Notice inviting Tender, shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury. |

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| | | 19.4 All compensation or other sums of money payable by the contractor to the Government under the terms of this contract or under any other contract with Government may be deducted from, or paid by the sale of a sufficient part of the performance Security or from the interest arising there from or from any sums which may be due or become due to the contractor by the Government or any account whatsoever and in the event of his Performance Security being reduced by reason of any such deduction, or sale as aforesaid, the contractor shall within ten days thereafter make good in cash or securities, endorsed as aforesaid, any sum or sums which may have been deducted from or realized by the sale of his Performance Security or any part thereof.<br><br>Government shall not be responsible for any loss of securities or any depreciation in the value of securities while in their charge nor for loss of interest thereon.<br><br>19.5 In the event of contract being cancelled, under Condition 52, 53 & 54 of General Conditions of Contract, the Performance Security shall be forfeited in full and shall be credited into Consolidated Fund of India. |
| 48 | 18 | **Condition 46 (Inspection of the work), Second Sub Para, Line 1 To 5:-**<br>**FOR:-**"Should the G.E. consider, at any time during the construction or reconstruction or prior to the expiration of period of a twelve calendar months after the works have been handed over to Government (hereinafter referred to as the "defects liability period") that any work has been executed with unsound, imperfect or unskillful workmanship or of a quality inferior to that contracted for or not otherwise in accordance with the Contract….."<br><br>**READ:-"**Should the GE consider, at any time during the construction or reconstruction or prior to the expiration of period of:-<br><br>(a) Thirty six calendar months after the works have been handed over to Government (herein after referred to as the "defects liability period") for Runway works, Marine & Harbour works, High Altitude works & Specialist works of Hospitals & Medical Equipment / Medical Gas.<br><br>(b) Twenty four calendar months after the works have been handed over to Government (herein after referred to as the "defects liability period") for works other than mentioned in sub para(a) above.<br>***(The period of defects liability period shall be specified in the tender documents.)***<br>That any work has been executed with unsound, imperfect or unskillful workmanship or of a quality inferior to that contracted for or not otherwise in accordance with the contract……" |
| 49 | 18 | **Condition 46 (Inspection of the work), Second sub Para, Line 1 To 5:-**<br>**FOR:- "**Should the GE consider, at any time during the construction or reconstruction or prior to the expiration of period of:-<br><br>(a) Thirty six calendar months after the works have been handed over to Government (herein after referred to as the "defects liability period") for Runway works, Marine & Harbour works, High Altitude works & Specialist works of Hospitals & Medical Equipment / Medical Gas.<br><br>(b) Twenty four calendar months after the works have been handed over to Government (herein after referred to as the "defects liability period") for works other than mentioned in sub para(a) above. |

**MILITARY ENGINEER SERVICES**
**AMENDMENT TO GENERAL CONDITIONS OF CONTRACT IAFW - 2249 (1989 PRINT)**

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|---|---|---|
| | | ***The period of defects liability period shall be specified in the tender documents.)***<br>That any work has been executed with unsound, imperfect or unskillful workmanship or of a quality inferior to that contracted for or not otherwise in accordance with the contract……"<br><br>**READ:-**"Should the G.E. consider, at any time during the construction of reconstruction or prior to the expiration of period of:-<br>(a) Thirty six calendar months after the works have been handed over to the Government (hereinafter referred to as the "defects liability period" ) for Runway works, Marine & Harbour works, High Altitude works & Specialist works of Hospital & Medical Equipment/ Medical Gas.<br>(b) Twelve calendar months after the works have been handed over to the Government (here-in-after referred to as the "defects liability period") for works of Periodical Services and Welcome Maintenance, Minor works and Revenue Works and Works for 'Maintenance & Operation' including Repairs and Works for Maintenance & Repair/Replacement'.<br>(c) Thirty days or date of taking over of the installation by MES, whichever is earlier, for works of Manning & Operation' & Maintenance & Operation' involving predominantly Outsourcing of service through Deployment of Manpower/Work-Force and not involving Repair/Replacement.<br>(d) Twenty four calendar months after the works have been handed over to the Government (hereinafter referred to as the "defects liability periods") for works other than mentioned in sub para (a) to (c) above."<br>***(The period of defects liability period shall be specified in the tender documents.)***<br>That any work has been executed with unsound, imperfect or unskillful workmanship or of a quality inferior to that contracted for or not otherwise in accordance with the contract……" |

**SIGNATURE OF THE CONTRACTOR** **DCWE (CONTRACTS)**
**DATED:** **FOR ACCEPTING OFFICER**

# Annexure 'A'

**Condition 70OF IAFW-2249.**

The existing descriptions of Condition 70 of IAFW-2249 shall be substituted by the revised description as under through amendment:

**ARBITRATION**

(a) **Arbitration Where Applicability of Section 12 (5) of the Arbitration and Conciliation Act Has Been Waived Off:-**

All disputes, between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them, be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to Rs 100 Crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs 100 Crore) from MoD Panel of Arbitrators. The officers so considered for appointment of Arbitrator, either as sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/Direct Final Examination of Sub-Division II of Institution of Surveyor (India) or similar other Institutes recognized by the Government of India.

In case of arbitration by Sole Arbitrator, the Arbitrator shall be appointed by the Authority mentioned in the contract document within a period of thirty days of having received the notice from any of the parties to Contract, out of MoD Panel of Arbitrators. In case of Arbitral Tribunal consisting of panel of three Arbitrators, both the parties will be asked by the Appointing Authority to suggest at least two names out of MoD Panel of Arbitrators within thirty days. The Appointing Authority will appoint two Arbitrators, one Arbitrator each out of the suggested names. The two Arbitrators so appointed will select one Arbitrator from the MoD Panel of Arbitrators who will be the 'Presiding Arbitrator'. The Serving Officer(s) so appointed as Arbitrator(s), either as Sole Arbitrator or as one of the three Arbitrators in the Arbitral Tribunal, can continue as Arbitrator even after retirement, provided both the parties to the Contract give written consent to this effect. In such case, however, the Arbitrator shall not be entitled for any fee even after retirement.

(b) **Arbitration Where Applicability of Section 12 (5) of Arbitration & Conciliation Act Has Not Been Waived Off**.

All disputes, between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them, be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to Rs 100 Crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs 100 Crore). The Officers so considered for appointment as Arbitrator, either as Sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/Direct Final Examination of Sub Division II of Institution of Surveyor (India) or similar other Institutes recognized by the Government of India.

In case of arbitration by sole Arbitrator, the Arbitrator shall be appointed by the authority mentioned in the contract document from the MoD Panel of Arbitrators within a period of thirty days of having received the notice from any of the parties to Contract.

In case of Arbitral Tribunal consisting of panel of three Arbitrators, both the parties shall be asked by the Appointing Authority to suggest at least two names out of MoD Panel of Arbitrators within thirty days. The Appointing Authority will appoint two Arbitrators, one Arbitrator each out of the suggested names. The two Arbitrators so appointed shall select one Arbitrator from the MoD Panel of Arbitrators who will be the 'Presiding Arbitrator'.

## Annexure 'A' (contd…….)

(c) **Common for All Arbitration**.

Unless both parties agree in writing, such reference shall not take place until after the completion or alleged completion of the works or termination or determination of the contract under Condition Nos 55, 56 and 57 hereof.

Provided that in the event of abandonment of the works or cancellation of the Contract under Condition No 52,53 or 54 hereof, such reference shall not take place until alternative arrangements have been finalised by the Government to get the works completed by or through any other Contractor or Contractors or Agency or Agencies.

Provided always that commencement or continuance of any arbitration proceeding hereunder or otherwise shall not in any manner militate against the Government's right of recovery from the Contractor as provided in Condition 67 hereof.

If the sole Arbitrator or one or more Arbitrators of the Arbitral Tribunal so appointed resign(s) from his/her appointment or vacate(s) his/her office or in unable or unwilling to act due to any reason whatsoever, the Authority appointing him/her will appoint a substitute Arbitrator to act in his/her place in the manner specified hereinabove. In case the Arbitrator resigning in this manner is the Presiding Arbitrator, the other two Arbitrators of the Arbitral Tribunal shall appoint the substitute Presiding Arbitrator.

The Arbitral Tribunal may proceed with the arbitration, exparte, if either party, in spite of a notice from the arbitrator fails to take part in the proceedings.

The Arbitral Tribunal may from time to time with the consent of the parties, enlarge the time for making and publishing the award subject to the limit laid down in the Arbitration & Conciliation Act 1996 as amended up to the date on which arbitration proceedings commence.

The Arbitral Tribunal shall make the award within the period as provided in the Arbitration & Conciliation Act 1996 (as amended up to the date on which arbitration proceedings commence) from the date of entering on the reference or within the extended period as the case may be on all matters referred to it and shall indicate findings along with sums awarded separately on each individual item of dispute. The Arbitral Tribunal shall give reason for the award in each and every case irrespective of the value of claims or counter claims.

The venue of Arbitration shall be such place or place or places as may by fixed by the Arbitral Tribunal in its sole discretion.

The Award of the Arbitral Tribunal shall be final and binding on both parties to the Contract.

**SIGNATURE OF THE CONTRACTOR** **DCWE (CONTRACTS)**
**DATED:** **FOR ACCEPTING OFFICER**

## SCHEDULE OF MINIMUM WAGES

1. It is hereby agreed that the "**Schedule of Minimum wages**" as published by Govt of India/State Govt/Govt local authorities whichever is highest and which specifies minimum rates of wages for various categories of workman as applicable on the last due date of receipt of this tender shall form part of these tender document.

The minimum rates of wages shall consist of all inclusive rates and include also the wages for weekly day of rest.

My/our signature here under amount to my/our having signed the aforesaid documents forming part of this tender. We also here by agree that payment to labours shall be made as per minimum wages notified by State Govt or Central Govt whichever is high.

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

## SPECIAL CONDITIONS AND REQUIREMENTS

### 1. GENERAL

1.1 The following Special Conditions shall be read in conjunction with the General Conditions of Contracts, IAFW-2249 and IAFW- 1779-A (Revised-1955) including errata/amendments thereto. Any provision in these special conditions if at variance with that of the aforesaid documents shall take precedence there over. The term General specification used in any of the documents forming part of the contract refers to the specification contained in the MES Schedule.

### 2. ADMISSION TO SITE BY CONTRACTOR AND RESPONSIBILITY TO ASCERTAIN HIS OWN INFORMATIONS

2.1 Refer paras 11 and 15 of Notice of Tender (IAFW-2162) and Condition 4 of IAFW-2249. The tenderer shall be deemed to have satisfied himself before submitting his tender as to the nature of the ground and sub-soil ( so far as it is practicable), the climatic conditions, the form and nature of the site, nature of work and the manner of access to the site and the accommodation he may require. In general, the tenderer shall be deemed to have himself obtained all necessary information on inspection of site, as to risks, contingencies and other circumstances which may influence or affect his tender. No extra payment consequent on any mistake or misunderstanding or otherwise on this account shall be allowed.

2.2 The tenderer shall be deemed to have visited the site(s) before quoting rate and made themselves familiar with the working conditions, whether he actually inspect the site(s) or not.

### 3. AVAILABILITY OF LAND FOR STORAGE OF MATERIALS, ACCOMMODATION FOR LABOUR ETC.

3.1 Delete the following from lines 5 to 9 of sub para 1 of Condition 24 of IAFW-2249 General Conditions of Contracts and insert as under in lieu thereof: -

"The contractor shall be permitted to store his materials including erecting temporary sheds thereof; temporary Workshops and the like at the area of land, marked for this purpose, on the site plan/shown by the GE free of rent, if it is on class A-1 land. For other types of land he shall be charged licence fee Re 1/- per year or part of a year for each and every separate area of land allotted to him. No land shall be allotted to the contractor for construction of huts for accommodation of labour. Before establishment of labour camps, the permission of the same shall be obtained from Stn HQ controlling the cantonment area.

### 4. CONDITIONS OF WORKING IN RESTRICTED AREA

4.1 Work under this contract lies in Restricted Area.

4.2 **VISIT TO THE SITE BEFORE TENDERING**: - Permission to enter the Restricted Area at the time of submission of tenders can be obtained through the GE/ Factory. Tenderers are advised to send prior intimation to the GE/ Factory about the particulars of their agents, representatives, etc. if any, dates any time of their proposed visits so that necessary arrangements may be made by GE/ Factory to secure admission whether a tenderer visits the site or not shall be deemed to have full knowledge of the restrictions on entering into/ exist from and working within the Restricted Area.

4.3 **ENTRY/EXIT** :- The contractor, his agents(s), representatives, workmen, etc. and his materials, carts trucks or other means of transports etc. will be allowed to enter through and leave from only such gate or/ gates and at such times as the GE or authority incharge of the Restricted Area may at their sole discretion permit to be used. The contractor's authorized representative is required to be present at the place of entry and exit for the purpose of identifying his carts, trucks, etc. to the personnel incharge of the security of Restricted Area.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

4.4 **IDENTITY CARDS OR PASSES:**-The contractor, his agents and representatives are required individually to be in possession of identity card or pass duly verified by the Police Department. The identity card or pass will be examined by the security staff at the time of entry into or exit from the Restricted Area and also at any time or number of times inside the Restricted Area.

4.5 **IDENTITY OF WORKMEN**:-

4.5.1 Every workman shall be in possession of an identity card. The identity will be issued after a thorough investigation of the antecedents of the labourers by the contractor and attested by Office-in-Charge of the unit concerned in accordance with the standing rules and regulations of the unit.

4.5.2 Contractor shall be responsible for the conduct and action or his workmen, agents or representatives.

4.6 **SEARCH**. Though search of all persons and transport shall be carried out at each gate and for as much time as a gate is used for entry or exit and may also be carried out at any time or any number of times at work site with the Restricted Area.

4.7 **FEMALE SEARCHES** :- If contractor desires to employ female labour on works to be carried out inside the area of a factory, Depot, Park etc and female searcher is not borne on the authorized strength of the factory, Depot Park, etc. at the time of submission of tender, he shall be deemed to have allowed in his tender for pay and allowances etc for a female searcher (class IV/Group 'D') servant calculated for the period of female labour is employed by him inside the area. If more than one contractor has/ have to be employed in addition to the additional female searchers (s) shall be distributed on an equitable basis between the contractors employing female labour taking into consideration the value and period of completion of their contracts. The GE's decision in regard to the amount recoverable on this account from any contractor shall be final and binding.

4.8 **WORKING HOURS**:-

4.8.1 The units controlling restricted area, usually work during six days in a week and remain closed on the 7th day. The working hours available to the contractor's labour and staff are however, appreciably reduced because of the time taken in security checks observed at the time of entry, exit and during working hours.

4.8.2 The exact working hours, days and non-working days observed for the restricted area, where works are to be carried out, shall be deemed to have been ascertained by contractor before submitting his tender. The tenderers attention is invited to the fact that the total number of working hours for a unit is prescribed in regulations and they cannot be increased by the Garrison Engineer/ Factory.

4.8.3 Contractor's materials, transport etc shall normally be permitted to come in/go out of the area between 8 AM to 6 PM only or as directed by the authority.

4.8.4 Contractor may also be allowed to carry out the work beyond 6.00 PM and up to 8.00 PM (day and night). However no movements of materials and transport to/out of site of works shall be permitted during night unless special permission is obtained from the unit authority.

4.9 **WORK ON HOLIDAYS: -** The contractor shall not carry out any work on Gazetted Holidays, weekly holidays and other non-working days except when he is specially authorized to do so by GE/ Factory. The GE may at his sole discretion declare any day as holiday or non-working day without assigning any reason for such declaration.

4.10 **ACCESS TO RESTRICTED AREA AFTER COMPLETION OF WORKS**: - After the works are completed and surplus stores etc., returned to the contractor, his agents, representatives or workmen etc. may not be allowed to have access to the restricted areas except for attending any rectification of defects pointed out to him by the GE/ Engineer-in-Charge.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

5. **CONTRACTOR'S REPRESENTATIVES AND WORKMEN**

5.1 Refer Condition 25 of IAFW -2249. The contractor shall employ only Indian Nationals as his representatives, servants and workmen and verify their antecedents and loyalty before employing them for the works. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with works. If for reasons of technical collaboration, or other consideration employment of any foreign national is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of his tender.

6. **SECURITY OF CLASSIFIED DOCUMENTS**.

6.1 Contractor's special attention is drawn to Conditions 2-A and 3 of IAFW-2249 (General Conditions of Contracts). The contractor shall not communicate any classified information regarding the work either to subcontractors or others without the prior approval of the Engineer-in-Charge. The contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of the work, and shall return all documents on completion of the works or earlier on determination/Termination/cancellation of the contract. The contractor shall along with the final bill, attach a receipt of his having returned the classified documents as per condition 3 of IAFW-2249 (General Conditions of Contracts).

7. **MINIMUM WAGES PAYABLE**.

7.1 Refer Condition 58 of IAFW-2249. The contractor shall not pay wages lower than minimum wages for labour as fixed by the Govt of India/State Govt/Union Territory under Minimum Wages Act or Contract Labour (Abolition and Regulation Act) whichever is higher.

7.2 The fair wage referred to in condition 58 of IAFW-2249 shall be deemed to be the same as the minimum wages payable as referred to above.

7.3. The contractor shall have no claim whatsoever, if on account of local factors and/or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

8. **CO-OPERATION WITH OTHER AGENCIES**

8.1 The contractor shall permit free access and generally afford reasonable facilities to other agencies or departmental workmen engaged by the Govt to carry out their part of the work, if any, under separate arrangements. The contractor shall not be allowed any extra payment on this account.

9. **QUARRIES**

9.1 Quarries are not available on land which is in the charge of the MES authorities. Condition 14 of IAFW-2249 shall thus be treated as deleted.

10. **WATER**

10.1 Water will be supplied by MES to the Contractor at Points shown on Site Plan/ as shown by the Engineer-in-Charge from Piped System and shall be paid by the contractor @ **Rs. 3.75 per Rs. 1000/-** worth of work done priced at contract rates. The Contractor shall arrange at his own expense for Storage of Water and Litting Pumping, carrying of conveying Water to the Site of Work as required. In case Water Points are not marked on the Site Plan, the Water shall be supplied at ONE POINT at decided by Engineer-in-Charge.

10.2 The supply of water may not be continuous. The contractor shall be deemed to have ascertained the hours of availability of water before submitting his tender. The MES do not guarantee the continuity of water supply and no compensation shall be allowed for intermittent or inadequate water supply and break down in the system.

10.3 The contractor, if he so desires, will be allowed to Decent the water supply by boring wells at his own cost at site(s) approved by the GE on completion of the work, the contractor shall remove pipes etc installed and make good the site as directed by Engineer-in-Charge without any extra cost to the Government. The contractor shall, however, ensure that water drawn from such sources is clean and free from all impurities and is suitable for mixing of concrete, mortar, washing, aggregate and curing of concrete as specified in IS-456.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

11. **SUPPLY OF ELECTRICITY**.

11.1 In case of contractor desires to buy electricity from MES he shall be charged for the Electric energy consumed at **Rs. 17.35** per unit for lighting and power.

11.2 Electric supply required for the work shall be made available by the MES authorities. KWH Meter to register the electric energy, main switch shall be provided and installed by the contractor. The contractor shall also provide all necessary cables fittings etc, from the tapping point in order to ensure a proper and suitable supply of electricity for execution of work. The electric fittings, cable etc, shall be in accordance with IE Rules. The contractor shall apply for electric connection through GE, who will give necessary assistance to provide the electric supply.

11.3 MES do not guarantee continuity of supply and no compensation whatsoever shall be allowed for supply becoming inter–mitten or for breakdown in the system.

11.4 GE or his representative shall be free to inspect all the power consuming devices or any electric lines provided by the contractor. Any devices or electric lines provided by the contractor, which is not to the satisfaction of the GE shall be disconnected from the supply, if so directed by him.

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12. **SAMPLE OF MATERIALS**

12.1 Refer condition 10 of I.A.F.W. 2249 and Clause No.1.6 & 1.7 of MES Sch Part I.

12.2 The tenderers are advised to inspect sample of the materials which are displayed in the office of the Garrison Engineer before submitting his tender. The tenderers shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials, he is required to incorporate in the work, irrespective of whether he has actually inspected or not. The materials to be incorporated in the work by the contractor shall conform to or shall be superior in quality to the sample displayed and shall comply with the specifications given hereinafter.

12.3 The contractor shall not procure materials unless the samples are first approved by the Garrison Engineer.

12.4 The materials other than materials of local origins shall comply with the requirement of the latest IS.

12.5 The cost of testing of material shall be borne by the contractor while quoting their lumpsum. This aspect shall be kept in mind and nothing extra shall be admissible on this account.

13. **RECORD OF MATERIALS.**

13.1 The quantity of materials such as paints, water proofing compound, chemicals for antitermite treatment and the like, as directed by the Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

13.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in measurement book shall be suitably marked for identification.

13.3 The contractor shall, on demand, produce to the GE, and original receipted vouchers/invoices in respect of the supplies. The vouchers/invoices so produced shall be defaced and stamped by Engineer-in-Charge indicating contract number, name of work, under his dated signature. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing, bearing manufacturer's marking except in the case of the requirement of material(s) being less than smallest packing.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

14. **PHOTOGRAPHS**

14.1 The contractor shall provide minimum 50 (fifty) photographs (coloured) in duplicate with negative of the important stages of the work in progress as directed by GE/ Engg-in-charge. The contractor shall arrange the equipment or any other attachment for taking the photograph.

14.2 The photograph shall be postcard size and shall be sharp, neat and clean and of standard as approved by GE. Cost of photographs shall be borne by the contractor.

14.3 In case photography is prohibited in this area, the photographs shall not be taken and no price adjustment shall be made on this account. The contractor shall not allow carrying the camera or the equipment connected with the photographs in the area/buildings without permission of the Engineer-in-Charge.

15. **VALIDITY OF TENDER**

15.1 The tender shall remain open for acceptance for a period of **60 (Sixty) days** from the due date of its submission excluding the date of submission of tender.

16. **CPM NET WOPK DIAGRAM**

16.1 The time and progress chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) shall consist of detailed net work analysis and a time schedule. The critical path net work shall be drawn jointly by the Engineer-in-Charge/GE and the contractor soon after acceptance of tender. The time scheduling of the activities shall be done by the contractor so as to finish the work within the stipulated time. On completion of the time schedule a firm calendar date schedule shall be prepared and submitted by the contractor to the Engineer-in-Charge/GE who shall approve it after due scrutiny. The schedule shall be submitted in four copies within two weeks from the date of handing over the site. In case the contractor fails, to submit the CPM net work diagram, the network prepared by the Engineer-in-Charge shall be binding on him.

16.2 During the currency of the work, the contractor is expected to adhere to the time schedule and this adherence shall be a part of his/their performance under the contract. During the execution of the work, the contractor is expected to participate in the reviews and updating of the net work undertaken by the Engineer-in-Charge/GE. These reviews may be undertaken at the discretion of the Engineer-in-Charge/GE, either as a periodic appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or otherwise. Any revision of the time schedule as a result of the review, shall be submitted within a week by the contractor to the Engineer-in-Charge/GE for his approval after due scrutiny.

16.3 The contractor shall adhere to the revised time schedule thereafter. In case of contractor disagreeing with revised schedule, the same shall be referred to the Accepting Officer whose decision shall be final, conclusive and binding. Engineer-in-Charge's approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW-2249 and separately regulated.

16.4 Contractor is expected to mobilise and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety.

16.5 No additional payment will be made to contractor for any multiple shift work or other incentive methods contemplated by him in his work schedule, even though the time schedule is approved by the department.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

17. **BLANK**

18. **SITE FOR EXECUTION WORK**

18.1 Site for execution of work shall be available as soon as the work is commenced on the date as mentioned in the work order No 1. However, in case due to unavoidable circumstances it is not possible to make the entire sites available on the date of the commencement, the contractor shall have to arrange his programme accordingly. No claim whatsoever for not getting the entire site on the date of commencement of work & for giving the site gradually shall be tenable.

19. **RECOVERY CHARGES FOR TESTING OF MATERIALS**

19.1 Where testing of contractor's materials is carried out in zonal/National test house/SEMT wing/Engineering college by the department, the contractor shall be liable to pay the testing charges as stated in Appendix `D' to particular specifications and shall be recovered from the contractor's running payment/final bill as the case may be.

20. **SAFETY PRECAUTION**

20.1 The contractor shall take every/precaution to control traffication road keeping danger boards, necessary lighting arrangements, fencing and watchman to avoid any damage. In case due to excavation or others the road is to be blocked the contractor shall, without any extra cost to the Govt. provide separate bypass so that normal traffic is not disturb.

20.2 Any damages to the existing road/building etc, shall be made good by the contractor with the same specifications as per existing work, without any extra cost to the Govt. In the event of contractor not fully complying with the above provisions to the satisfaction of the Engineer-in-Charge, the Engineer-in-Charge may provide the same for which the expenses incurred shall be recovered from the contractor.

21. **RECORD OF CONSUMPTION OF CEMENT**

21.1 The contractor shall maintain a pucca bound register with serially numbered pages with all pages initialed by Engineer-in-Charge against numbering showing quantities of cement records and used in the work daily and balance at the end of each day. The form of record shall be as approved by Engineer-in-Charge. The register shall be signed daily by the representative of MES and the contractor in token of verification of its correctness and shall be checked by Engineer-in-Charge, at least once a week and on the days cement is issued to the contractor.

21.2 The register shall be kept at site safe custody of the contractor representative during the progress of the work and shall on demand, be produced for verification to the inspecting officers.

21.3 On completion of the work the contractor shall deposit the cement register with the Engineer-in-Charge for record.

22. **BLANK**

23. **RETENTION MONEY/COMPENSATION FOR DELAY**

23.1 Refer condition 64 of IAFW-2249 (Advances on account) and Condition 50 of IAFW-2249 (Compensation for Delay).

23.2 For the purpose of calculating retention money and liquidated damages under condition 64 of IAFW-2249 and compensation for delay in completion of work under condition 50 of IAFW-2249, the value of contract as revised by above mentioned price variation under modified condition 63 of IAFW -2249 shall be taken into account.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

24. **OFFICIAL SECRET ACT**

24.1 Refer Condition 24 of General Condition of Contract IAFW-2249. The contractors attention is invited to Indian Official Secret Act 1923 (xxx of 1923) particularly Sec 5 thereof. The contractors shall be bound by the provision of this Act.

25. **DAMAGE TO EXISTING STRUCTURE/BUILDING**

25.1 Any damage caused to the existing structure during the execution of work shall be made good by the contractor at his own cost and the site of work left clean and tidy on completion. Rectifications/ making good etc. shall conform to the standard of materials originally used in the work and finished work shall match with existing work in all respects to the entire satisfaction of the Engineer-in-Charge. In case of any dispute on this account the matter shall be referred to the GE whose decision in writing shall be final, conclusive and binding.

26. **ROCK MET IN EXCAVATIONS**

26.1 If during excavation, rock is met with GE shall be informed of this in writing by the contractor. The nature/type of rock met with shall be decided by the GE, whose decision is final and binding. If the nature of rock met with is `Hard Rock' the same shall be priced and paid as excavation in hard rock at the rates as applicable to this contract. Hard rock obtained from excavation shall be properly stacked, measured and recorded in measurement book and will be the property of the Contractor. Credit at the rate of Rs 750.00 per cubic metre of hard rock obtained shall be recovered from the contractor. The measurement shall be signed by Engineer-in-Charge and contractor. Hard rock obtained thus can be reused in the execution under this contract subject to fulfillment of requisite specification specified here-in-after and after written approval of Garrison Engineer.

27. **BLANK**

28 **RE-IMBURSEMENT/REFUND ON VARIATION IN "TAXES DIRECTLY RELATED TOCONTRACT VALUE**

(a) The rates quoted by the contractor shall be deemed to be inclusive of all taxes/cessesviz GST, duties, Royalities, Octroi& other levies payable under the respective statutes. No other levies, and/or imposition/abolition of any new/existing taxes/cesses, duties, Royalities, Octroi& other levies shall be made except as provided in para (b) here-in-below.

(b) (i) The taxes/cesses which are levied by Govt at certain percentages rates of contract sum/Amount shall be termed as "taxes directly related to contract value" such as GST, Labour welfare cess/tax and like as applicable but excluding Income Tax. The tendered rates shall be deemed to, be inclusive of all "taxes directly related to contract value" with existing percentage rates as prevailing on last due date for receipt of tenders. Any increase in percentage rates of "taxes directly related to contract value" with reference to prevailing rates on last due date for receipt of tenders shall be reimbursed to the contractor and any decrease in percentage rate of "taxes directly related to contract value" with reference to prevailing rates on last due date for receipt of tenders shall be refunded by the contractor to the Govt/deducted by Govt from any refunded due to the contractor. Similarly imposition of any new "taxes directly related to contract value" after the last due date for receipt of tenders shall be reimbursed to the contractor and abolition of any "taxes directly related to contract value" prevailing on last due date for receipt of tenders shall be refunded by the contractor to the Govt/deducted by the Govt from the payments due to the contractor.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

(ii) The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "taxes directly related to contract values", give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating thereto which he may be in a position to supply. The contractor shall submit the other documentary proof/information's as the GE may required.

(iii) The contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorised representative of Govt, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other informations as the GE may require.

(iv) Reimbursement for increase in percentage rates/imposition of "taxes directly related to contractor value" shall be made only if the contractor necessarily & properly pay additional "taxes directly related to contract value" to the Govt, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt Authority and submits documentary proof for the same as the GE may require

29. **TESTING OF MATERIALS**:

29.1 Refer condition 10 of IAFW-2249. All tests shall be got carried out by the contractor from laboratories / testing agencies.

29.2 The cost of materials, labour to be consumed in testing, transportation charges, testing charges and like shall be borne by the contractor.

29.3 The contractor may however get the materials tested from Zonal laboratory (where available) for which the testing charges shall be borne by the contractor. In case of non-availability of facilities for such tests in the Zonal laboratory, the test may be carried out in National Test House/ SEMT wing / Engineering Colleges/ Govt recognized laboratories as approved by GE.

30. **TRAINING OF MES STAFF**

30.1 The contractor shall give training to MES staff as detailed by the Engineer-in-Charge for 15 days period regarding operation, running and maintenance of various equipment, installed by him, as desired by the Engineer-in-Charge. Unit rates lump sum quoted by the contractor shall be deemed to have included the rates for the same and no extra payment shall be made to the contractor on this account.

31. **CLEANING DOWN**

31.1 Refer condition 49 of IAFW-2249 (General conditions of contracts).

31.2 The contractor shall clean all floor, walls, remove cement/lime/paint marks/drops, etc clean the joinery glass panes etc, touch up all painters work and carryout all other necessary items of work in connection herewith and leave the whole premises clean and tidy before handing over the building(s).

32. **SCOPE OF CONCILIATION**

32.1 The scope of conciliation shall be restricted to the following types of disputes with financial limit as indicated therein: -

(a) Disputes relating to levy of compensation for delay in completion actual amount of compensation.

(b) Disputes relating to technical examination of works.

(c) Disputes relating to interpretation of the provisions of the contract with reference to their application to parties.

(d) Disputes relating to non-return of schedule `B' stores over-issued to contractor.

(e) Any other dispute having fair chances of being resolved by conciliation and considered fit to be referred to conciliation by the parties.

For item (b), (c), (d) and (e) each as stated above the financial limit shall be Rupees two lakhs or one percent of the contract amount whichever is less.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

32.2 **COMMENCEMENT OF CONCILIATION PROCEEDINGS**

32.2.1 The party initiating conciliation shall send to the other party a written invitation to conciliate, briefly identifying the subject of the dispute.

32.2.2 Conciliation proceedings shall commence when the other party accepts in writing the invitation to conciliate.

32.2.3 If the other party rejects the invitation, there will be no conciliation proceedings. If the party initiating conciliation does not receive a reply within 30 days from the date on which he sends or within such other periods of time as specified in the invitation, he may elect to treat this as a rejection of the invitation to conciliate and if he so elects, he shall inform in writing the other party accordingly.

32.3 **NUMBERS OF CONCILIATORS**

32.3.1 There shall be a sole conciliator.

32.4 **APPOINTMENT OF CONCILIATOR**: -

32.4.1 All disputes brought out in para 38.1(a) to (e) herein before shall be referred to the sole conciliator viz serving officer not below the rank of Superintending Engineer/Superintending Engineer (QS & C) having degree in Engineering or equivalent or having passed final/direct final examination of sub division II of Institution of Surveyors (India) to be appointed by the Engineer-in-Chief, Army Headquarters, New Delhi or in his absence the officer officiating as Engineer-in-Chief or Director General of works specifically delegated by the Engineer-in-Chief in writing.

33. **STATUS OF EFFECT OF SETTLEMENT AGREEMENT**: -

33.1 The settlement agreement signed by the parties as a result of conciliation proceedings shall have the same status and effect as it is an arbitral award on agreed terms.

34. **HANDING OVER OF SITE**: - Site for execution of work will be available as soon as the work is awarded. In case it is not possible to make the entire site available on the award of work, the contractor will have to arrange his working program accordingly. No claim whatsoever, for not giving entire site on award of work and for giving site gradually, will be tenable.

**35. OUTPUT OF POWER ROAD ROLLER:**

(a) Reference Condition 15 of IAFW-2249.

(b) Where road roller(s) is/are hired by the department to the contractor, a log book for each road roller shall be maintained by the department for recording hours of working of the road roller. In case, however, when the contractor procures road roller(s) from sources other than the department, a logbook for each road roller shall be maintained by him for recording hours of working of the road roller. Entries in the logbook shall be signed by the contractor or his authorised representative and by the Engineer-in-Charge at the site of work.

(c) To ensure proper consolidation, roller must work for at least number of days assessed on the basis of output given here under. If the roller has not worked for the number of days so assessed, recovery shall be effected from the contractor for the number of days falling short of the days assessed on the basis of output stipulated. The recovery shall be affected as under:-

(i) Where road roller is hired out only by the department to the contractor, at rates given in Sch `C'.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

(ii) Where road roller is hired by the contractor only from sources other than the department, Rs.1500/- per working day of 8 hours for 8 to 10 ton power roller.

(iii) Where road roller is hired by the contractor from the department and also from sources other than the department, at higher of the two rates given in Schedule `C' of contract and clause 25.3(b) above.

(iv) The above provision shall not, however, absolve the contractor of his responsibility of properly consolidating surface as required under the provisions of the contract.

(v) Output per day of 8 hours work of power roller (8 to10 Tonne):-

| | | | |
|---|---|---|---|
| (ca) | Consolidation of formation surfaces/sub-grade | - | 1850 Sqm |
| (cb) | Consolidation of stone soling 20 to 23 cm thick | - | 518 Sqm |
| (cc) | -ditto- but for 15 cm thick, | - | 800 Sqm |
| (cd) | Consolidation of water bound macadam (Stone metal) 7.5 cm consolidated thickness including spreading and consolidation with binding materials | - | 248 Sqm |
| (ce) | Consolidation of water bound macadam (Stone metal) 10 cm consolidated thickness including spreading and consolidation with binding materials. | - | 175 Sqm |
| (cf) | Consolidation of one coat surface dressing | - | 774 Sqm |
| (cg) | Consolidation of two coat surface dressing | - | 558 Sqm |
| (ch) | Consolidation of 2.5 cm thick premixed carpet including seal coat | - | 600 Sqm |
| (ci) | Consolidation of 2 cm thick premixed carpet including seal coat | - | 744 Sqm |
| (cj) | Consolidation of bituminous mixture, 2 parts of broken stone metal and one part of sand and bitumen, consolidated thickness 4 cms | - | 370 Sqm |
| (ck) | Consolidation of 15 cm thick earth berm | - | 1800 Sqm |
| (cl) | Consolidation of broken stone base | - | 500 Sqm |
| (cm) | Granular sub base with vibratory roller | - | 50 cum/hour |
| (cn) | Wet mix macadam with vibratory roller | - | 60 cum/hour |

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

36. **TESTING OF MATERIALS**

36.1 **A LEVEL TESTS FOR WORKS COSTING 100 LAKHS AND ABOVE**: The contractor shall set up site laboratory for testing of materials (except Sch "B" materials) for "A" level tests as listed in Appx "C". The contractor shall arrange all equipment/ machines for the tests specified in Appx "C" as A level tests at his own cost with prior approval of GE. This cost shall be included in the lump-sum costs quoted by the contractor. The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Engineer-in-charge. The successful tests result thereof shall be recorded and signed jointly by the contractor and the Eng-in-charge. The charges for these tests i.e. A level tests carried out in site laboratory of the contractor shall not be recovered. In case, the contractor has not set up the site laboratory and the tests are carried out in zonal or any other laboratory set up/approved by GE, the recovery shall be made at rates applicable i.e. as given in Appx "C".

36.1.1 **A LEVEL TESTS FOR WORKS COSTING UPTO 100 LAKHS**: The contractor may set up site laboratory at his option for works costing upto 100 lakhs. The other stipulations will be same as specified in preceding para 29.1. However in case the contractor has not set up the site laboratory and tests are carried out in zonal or any other laboratory approved/set up by GE/AGE(I), the recovery shall be made at rates applicable ie as given in Appx "C".

36.2 **B & C LEVEL TESTS**: For tests of "B" & "C" level as indicated in Appx "C" , the contractor shall provide all facilities for testing of materials at CTL Siliguri or after getting NAC from CTL and getting it done from approved test house/Engineering College at his own cost. The lump sum/rates quoted by the contractor shall deemed to be inclusive of these tests. The rates of various tests conducted in laboratory of MES are indicated in Appx "C". The contractor shall bear the actual charges of "C" level tests (to be done in labs other than MES labs) irrespective of rates indicated in Appx "C". Wherever it is convenient to get „B" level test done at approved test house/Engg. College, the same can be done at the cost of the contractor and no separate recoveries will be made by the Department for the same.

36.3 **INDEPENDENT TESTING / THIRD PARTY TESTING:** The further tests listed in table hereinafter at **Appendix- "D"** will be carried out in site lab and independent testing in IITs/SEMT/CRRI/NIT/National Test House/Govt. Engineering Colleges through department as mentioned against each. In case method of tests is at variance with the classes mentioned hereinbefore the relevant IS shall supersede in this regard. The charges for getting the tests done in independent lab through department including transporting the testing material shall be deemed to be included in the rates quoted and no extra claim on this account shall be admissible.

36.4 The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective tests or whenever the test is due whichever is earlier.

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

**ANNEXURE-IV**

**(This shall form part of Contract Agreement)**

**FORMAT FOR QUALITY CONTROL PLAN**
**(To be submitted by Contractor within 30 days of commencement of Contract)**
**PART - I**

1. Contract Agreement Reference No. :

2. CPM network prepared and approved by GE (I) :

3. Resource scheduling done based on CPM :

4. Site Laboratory (with equipments) set up as per Contract Agreement (CA) :

5. Concrete mix design submitted and approved :

6. Preliminary works completed to standard engineering practice :

7. Arrangements for water made (including testing of water) :

8. Arrangement for electric supply made :

9. Materials

| Srl No. | Item | Source as per CA | Contractor's plan of sourcing | Refer to testing clause | Agency for testing | Responsibility |
|---|---|---|---|---|---|---|
| | | | | | | |

10. List of all T&P, make and numbers that the contractor would deploy at site of work. :
11. Name of person nominated by contractor for exercising quality control. :

12. Qualifications / Experience of person at Serial No. 11 above. :
13. Names of supervisors with their qualifications / experience employed by contractor. :

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

**ANNEXURE-IV (Contd/.....)**

**Format For Quality Control Plan (Contd/.....)**

14. Confirmation that contract requirement relating to quality of all materials and quality standards of workmanship and finishes and acceptance criteria are explained and understood by all. :

15. Confirmation that requirement of tests to be conducted on materials before approval of sample and during execution, test on workmanship, tests before acceptance including the testing procedure, sampling techniques, frequency and agencies responsible for testing are understood and shall be complied with. :

16. Method to be adopted for maintaining records of test result. :

17. Certificate that contractor shall maintain a log of all materials received at site as per the following format :- :

| Srl No. | Date | Material | Quantity received | Source | Whether as per approved sample or not | Tests carried out by supplier | Tests to be carried out before incorporation |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

18. General Remarks by contractor of his plan of actions to ensure that quality standards. :

**DATED :** **(SIGNATURE OF CONTRACTOR)**

**SPECIAL CONDITIONS AND REQUIREMENTS CONDTS.....**

**ANNEXURE-IV (Contd/.....)**

**Format For Quality Control Plan (Contd/.....)**

**PART – II**

**(To be completed by GE before forwarding for approval by CWE)**

1. Verification of Serial No. 2 to 8 of Part-I :

2. Verification of Serial No. 9 to 18 of Part-I :

3. Confirmation that Stage Passing Register laying down the stages and authority responsible for approving the same has been prepared, shown to contractor and kept at site. :

4. Confirmation that all sites as require by contract had been handed over to contractor on the date fixed in Work Order No. 1 :

5. Confirmation that arrangements for Govt. liability in supply of water and electricity has been made and no hold up on this account in expected. :

**DATED :**

**(SIGNATURE OF GE)**

**APPROVED BY CWE**

**NAME OF WORK: REPAIR/MAINTENANCE INCLUDING RE-CARPETING OF ROADS IN OMQ & SMQ AREA UNDER GE (AF) HASIMARA.**

## PARTICULAR SPECIFICATIONS

1. **GENERAL**

1.1 Work under this contract shall be carried out in accordance with Sch `A', particular specifications, and general specifications and other provisions in MES Standard Schedule of Rates (here-in-after called as MES Schedule or SSR) Part I and Part II read in conjunction with each other.

1.2 General Specifications referred to/mentioned in General Conditions of Contracts IAFW-2249 (1989- Print) and in other documents mean specifications including other provisions given in MES Schedule Part-I (2009) and Part-II (2020) including amendments & errata as applicable.

1.3 Materials & Workmanship required for these Works/Services shall be as described against 'Specifications & Workmanships' in the various Trade/Sections of the MES Schedule duly modified by these Particular Specifications here-in-after.

1.4 Reference to some paragraphs of MES Schedule has been made in these Particular specifications, but other paragraphs and provisions as applicable are also to be followed e.g. reference Clause No., pertaining to general workmanship for brick masonry, iron and steel work, etc. have not been made but provisions therein as required for the work are applicable.

1.5 Provisions contained in the MES Schedule Part II in the Preambles to the relevant Items of SCHEDULE 'A' shall also be read in conjunction with the Provisions contained in these documents.

1.6 Where specifications for any item of work are not given in MES Schedule or in these particular specifications, specifications given in relevant Indian standard specifications or code of practice shall be followed.

1.7 Where specifications/provisions for any item of work given in these particular specifications are at variance with the provisions/specification given in MES Schedule, specifications/provisions given in these particular specifications shall be followed. However for pricing of Deviation Orders, rates given in MES Schedule Part II shall be adopted.

1.8 Particular Specifications given here-in-after shall be read in conjunction with the Provisions in the MES Schedule and relevant Indian Standards referred to thereto and in these Particular Specifications. In case of any discrepancy, the Provisions in these Particular Specifications shall take precedence.

1.9 Materials/Accessories/Equipments for which MAKE have not been specified in these Particular Specifications, shall be of Standard Makes and shall strictly comply with Current Appropriate IS Specifications, for which IS Specification has not been issued/available, they shall comply with the Currents BS Specifications or as approved by GE.

1.10 These Particular Specifications shall deem to be includes the APPENDICES attached here-in-after.

2. **SCOPE OF WORK**

2.1 This contract includes for the full, final and entire completion of works described in Schedule `A' /BOQ. Tenderers are advised to visit the SITE(S) and ascertain for themselves the exact scope of Work and its Working Conditions, viz Working Hours, Availability of Site(s), etc. and quote their Tenders accordingly. Any Claim whatsoever on such/these accounts will not be entertained by the Government at a later Date/Stage.

**2.2 VISIT OF SITE**

2.2.1 The Tenderers are advised to Contact Engineer-in-Charge on any Working Day during Working Hours, who will arrange for their Visit to Site of Work. The Tenderer shall be deemed to have visited the Site of Work before Quoting their UNIT RATES and no Claim for any EXTRA PAYMENT on account of any misunderstanding, etc. will be entertained by the Department, irrespective of the fact whether they have actually visited the Site or not.

**PARTICULAR SPECIFICATIONS (CONTD…….)**

3. **MATERIALS**

3.1 All materials to be supplied by the contractor for incorporation in the work shall conform to relevant IS specifications. In case specification of any materials needed for incorporation in work is not contained in any of the contract documents, the specification of such materials/articles to be incorporated in work shall be got approved in writing from the GE before their incorporation in work by the contractor without any price adjustment.

3.1.1 The relevant Indian standards shall be of latest revision including amendments if any. The contractor shall incorporate in the work materials complying with the requirements of relevant Indian standards of the latest publication (Edition) including amendments/revision thereof without any price adjustment in the quoted rate.

3.2 As far as practicable all manufactured materials/articles other than those manufactured in contractor's workshop at site shall bear I.S. certification marks. In case, any article not bearing IS certification marks but conforming to relevant specification is proposed to be incorporated in the work, samples of the same shall be got approved in writing from GE before their incorporation in work. The contractor shall submit sufficient evidence to the GE to show that such articles conform to the relevant specifications and no adjustment shall be made on this account.

3.3 Materials/Accessories/Equipments for which MAKE have not been specified in these Particular Specifications, shall be of Standard Makes and shall strictly comply with Current Appropriate IS Specifications, for which IS Specification has not been issued/available, they shall comply with the Currents BIS Specifications or as approved by GE.

3.4 Manufactured materials/articles shall be brought at site in original sealed containers/packing bearing manufacturer's marking unless the quantity required is a fraction of smallest packing.

3.5 Materials such as bricks, aggregates etc, shall generally conform to the samples kept in Engineer-in-Charge's office in addition to their conformity with relevant specifications given in the tender documents. The sources and sample for all such materials shall be got approved from GE in writing before these materials are brought at site in bulk. The sources of some of the materials as applicable to the place of execution of the work are indicated as under:-

**Note:** Please note that the source given below are for guidance only, Contractor has to ascertain the availability of materials from the source given below before quoting their offer. In case of non-availability/scarcity of materials in the sources given below, no claim what-so-ever on this account shall be entertained by the Govt at later stage.

| | **MATERIALS** | | **SOURCE** |
|---|---|---|---|
| (a) | Sand coarse | : | Locally available as approved |
| (b) | Sand fine | : | -do- |
| (c) | Broken stone aggregates | : | Local approved stone crusher |
| (d) | Crushed stone aggregates | : | Local approved stone crusher |
| (e) | Bricks | : | Locally available as approved. |

3.6 Contractor at his own may procure materials from any other sources without any extra cost to the Department provided that the same conform to respective IS as stated here-in-before

3.7 Letters conveying approval of samples/materials by GE shall interalia mention sources of supply, name of manufacturer, trade name/brand (if applicable) and reference to clause of tender documents containing specifications of Particular materials.

3.8 Approval of Samples/Materials by GE shall in a Register labeled as 'SAMPLE APPROVAL REGISTER' to be maintained by the Engineer-in-Charge which interalia shall mention Sources of Supply, Name of Manufacturer, Trade Name/Brand (if applicable) and reference to Clause of Tender Documents containing Specifications of Particular Materials.

**PARTICULAR SPECIFICATIONS (CONTD.....)**

3.9 **STANDARD OF QUALITY & WORKSMANSHIP**

3.9.1 **GENERAL**

3.9.2 The Work shall strictly comply with the provisions contained in the Latest Edition of INDIA STANDARD CODE OF PRACTICE and/or IS SPECIFICATION as applicable, Works except where such Regulations and Rules are modified by these Particular Specifications.

3.9.3 All Works shall be carried out by properly Skilled Trademan. The Contractor shall on demand shall produce such evidence of Qualifications of his Workmen/Skilled Trademan/Supervisors/Engineers, either at the commencement of Work or at any time thereafter/during currency of the Contract. The entire Work shall be High Class with the best Workmanship and to the entire satisfaction of Engineer-in-Charge/ GE.

4. **SURFACE EXCAVATION**

4.1 Surface excavation shall be carried out as described in Para 3.3.2 of MES Schedule Part-I.

4.2 **SITE CLEARANCE AND SURFACE DRESSING**

4.2.1 Site clearance and surface dressing for the areas covered by road up to 3 meters both sides from the outermost execution details as per drawings for works shall be carried out as per paras 3.6 and 3.10 of MES Sch Part-I to the entire satisfaction of the Engineer-in-charge. All rubbish obtained from site clearance and all spoil obtained from surface dressing shall be removed, spread and leveled as directed by Engineer-in-charge.

4.3 **REMOVAL OF SURPLUS SOIL**

4.3.1 Surplus soil and/or rubbish obtained from excavation shall be disposed unless otherwise specified in particular specification here-in-after, upto a distance of 50 and not exc 500 meters at the location (s) as directed by Engineer-in-Charge, deposited, spread &leveled as specified in MES Schedule Part-I.

4.4. **HARD CORE**: - Refer Clause No. 3.27 and subsequent Clauses thereof of MES Schedule Part-I as applicable

4.4.1 The material for hard core shall be of hard broken stone or boulders, broken to gauge not exceeding 63 mm graded.

4.4.2 The hard core shall be spread, watered and rammed. The thickness of hard core shall be as specified in Schedule ‘A’/BOQ.

5. **REPAIR TO POT HOLES**

5.1 Refer Para 20.A.34.2 of MES Schedule Part-I. Potholes shall be repaired as described in BOQ and all as directed by the Engineer-in-Charge. Potholes repaired under this tender shall not be finished proud as mentioned in MES Schedule.

5.1.1 **POT HOLE GENERAL**

5.1.2 Refer to Para 20.A.21.12 of MES Schedule Part-I: Potholes repairs/carried out in the independent Locations shall be maintained a proud by at least 6.0mm to allow for subsequent settlement under Traffic. The Potholes (ruts) and other depressions should be drained off water and cut to regular shape with vertical side. All loose and disintegrated materials shall be removed and exposed surfaces swept and cleaned. Pot hole/Depression shall be then filled all as specified in respective items of Schedule 'A' and shall be compacted as directed by Engineer-in-Charge.

**PARTICULAR SPECIFICATIONS (CONTD.....)**

5.1.3 **POT HOLES REGISTER**: Pothole Register showing the number, size, location of Potholes is to be maintained which shall be signed jointly by Engineer-in-Charge and Contractor before execution of work.

5.1.4 **REPAIR TO POT HOLES**:-Refer Para 20.A.34.2 of MES Schedule Part-I.

5.2. **WATER BOUND MACADAM**

5.2.1 **Coarse aggregates: -** Refer Para 20.A.3.1 of MES Schedule Part-I. Coarse aggregate shall be crushed or broken stone aggregate and shall be hard, durable and generally free from flat, elongated, soft and disintegrated particles.

5.2.2 **Screenings: -** Refer Para 20.A.3.2 of MES Schedule Part-I. Screening shall be of hard stone and grading "B".

5.2.3 **Binding Materials: -** Refer Para 20.A.3.3 of MES Schedule Part-I.

5.2.4 **Workmanship: -** Refer para 20.A.21 and 20.A.21.1 to 20.A.21.13 of MES Schedule Part-I for workmanship.

5.2.5 Scarifying and Renewing of Water Bound Macadam surface: - Refer para 20.A.34.1 of MES Schedule Part-I

5.4 **PAVING BITUMEN (BINDER)**

5.4.1 The binder shall be paving bitumen. The grade of bitumen shall be VG-30 for premix carpet, SDAC, Bituminous Macadam and liquid seal coat conforming to IS-73. Tack coat shall be paving bitumen primer (Rs- 1)

5.4.2 **CHARACTERISTIC & REQUIREMENT OF PAVING BITUMEN (VG-30/ VG-10)**

5.4.2.1 Characteristic and Requirement of Paving Bitumen (IS: 73) as under:-

| Characteristics | Requirements for Paving Bitumen (VG – 30) | Requirements for Paving Bitumen (VG – 10) | Test Method |
|---|---|---|---|
| Flash Point, Cleveland Open Cop, $^0$C, Min | 220 | 220 | IS : 1202 |
| Softening Point, $^0$C, Min, R & B | 47 | 40 | IS : 1205 |
| Penetration at 25 $^o$C, 100gm, 5s, 1/10mm | 50-70 | 80-10 | IS : 1203 |
| Viscosity Ratio at 60 degree cen max | 4 | 4 | - |
| Ductility at 25$^o$C, cm, Min | 40 | 75 | IS : 1208 |
| Matter soluble in trichloroethylene, percent by mass, Min | 99 | 99 | IS : 1216 |
| Absolute Viscosity at 60$^o$C, Poise | 2400 | 800 | IS : 1206 |
| Kinematic Viscosity at 135$^o$C, CST,Min | 350 | 250 | IS : 1206 |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

5.4.3 The Bitumen shall be procured from the any of the following main producer:-

(i) M/s Indian Oil Corporation Ltd.
(ii) M/s Bharat Petroleum Corporation Ltd.
(iii) M/s Hindustan Petroleum Corporation Ltd.
(iv) Govt. Refineries

5.4.4 Bitumen shall be supplied in "Bulk", however, to ensure that no adulteration with original material takes place in transit, the contractor shall deploy proper watch and ward during transit of material. The contractor at his option may supply "Packed" bitumen without any extra cost to the Government.

5.4.5 The contractor shall submit original purchase vouchers along with manufacturers test certificate for the total quantity of bitumen supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the Garrison Engineer along with the relevant document. The original purchase vouchers and the test certificate shall be verified for subject contract and defaced by the Engineer-in-Charge and kept on record in the office of the Garrison Engineer duly authenticated and with cross reference to the consignment. The Accepting Officer may order a Board of Officers for random check of bitumen and verification of connected documents during the currency of contract. If the Garrison Engineer desires separate test of bitumen, the test shall be got done through a recognized and approved laboratory/institution as approved by Garrison Engineer at no extra cost to Government.

5.4.6 The bitumen brought at site shall be recorded in Measurement Book **"Not to be abstracted"** and shall be signed by both the contractor and Engineer-in-Charge.

5.4.7 **TESTING OF PAVING BITUMEN:** The contactor shall submit test certificates for each consignment brought at site. The material shall also be tested at site in the field laboratory at the stipulated frequency as 4.1, but cannot be performed at the field lab established by the contractor, shall be carried out at a laboratory as specified for Job Mix Formula, expenditure of such tests shall be borne by the contractor. In addition, GE may also get it tested independently from approved laboratory mentioned here-in-before. Contractor shall provide all facilities like material, transportation of materials to lab, etc. for testing by the Deptt. The test charges for such independent tests only shall be borne by the Deptt.

5.5 **SEMI-DENSE ASPHALTIC CONCRETE (SDAC)**:-

5.5.1 The Semi Dense Asphaltic Concrete shall consist of coarse aggregates, fine aggregates and filler in suitable proportions and mixed with sufficient binder. Every mix should be an economical blend of proper gradation of aggregates and adequate proportion of bitumen to fulfill the desired properties.

5.5.2 **AGGREGATE GRADING AND BINDER CONTENT**: The combined grading of fine, coarse aggregates and binder content shall conform to the following:-

| | Grading | |
|---|---|---|
| (i) | Nominal agg size | 13.2mm(Wearing Course) |
| (ii) | Thickness of layer | 30 to 40 mm |
| (iii) | Grade of Bitumen | Vg-30 |
| | **IS sieve(mm)** | **Percent passing** |
| | 26.5 | - |
| | 19 | 100 |
| | 13.2 | 90-100 |
| | 9.5 | 70-88 |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

| | | |
|---|---|---|
| | 4.75 | 53-71 |
| | 2.36 | 42-58 |
| | 1.18 | 34-48 |
| | 0.6 | 26-38 |
| | 0.3 | 18-28 |
| | 0.15 | 12-20 |
| | 0.075 | 4-10 |
| | Bitumen content % by mass of total mix | Min 5.4 |

**Note: -** 1. The nominal maximum particle size is the largest specified sieve size up on which any of the aggregate is retained.

2. Correspond to specific gravity of aggregates being 2.7. In case aggregates have specific gravity more than 2.7, the minimum bitumen content can be reduced proportionality. Further the region where highest daily mean air temperature is $30^0$C or lower and lowest daily air temperature is $-10^0$C or lower, the bitumen content may be increased by 0.5 percent.

5.5.3 **COARSE AGGREGATE**: Coarse aggregate shall be crushed material retained on IS 2.80mm sieve and shall be crushed stone from approved quarry and shall consist of angular fragments clear, tough and durable rock, free from disintegrated pieces and organic or other deleterious matters and adherent coatings. The aggregates shall not contain more than 6 percent by weight of flat/longed pieces (Flat piece is one having ratio of "width/thickness" of more than 4, elongated piece is where the ratio "length/width" is more than 4). The aggregates shall preferably be hydrophobic and of low porosity. If hydrophanous aggregates are to be used, which in normal circumstances shall be avoided, bitumen shall be treated with anti-stripping agents of appropriate quality in suitable doses. The aggregates shall satisfy the following physical requirements: -

| Property | Test | Specification | Frequency |
|---|---|---|---|
| Cleanliness (dust) | Grain size analysis (IS-2386 Part-I) | Max 5% passing 0.075mm sieve | 1 test for 100CUM |
| Particle shape | Combined Flakiness and elongation Indices (IS:2386 Part-I) | Max 35% | 1 test for 200CUM |
| Strength* | Los Angeles Abrasion Value (IS:2386 Part-IV)<br>*Aggregate Impact Value (IS:2386 Part-IV) | Max 30%<br>Max 24% | 1 test for 200CUM |
| Durability | Soundness either (IS:2386 Part-V):<br>Sodium Sulphate or<br>Magnesium Sulphate | <br>Max 12%<br>Max 18% | 1 test for 200CUM |
| Polishing | Polished Stone Value (BS : 812-114) | Min 55 | |
| Water Absorption | Water absorption (IS:2386 Part-III) | Max 2% | 1 test for 200CUM |
| Stripping | Coating and Stripping of Bitumen Aggregate Mix (IS:6241) | Minimum Retained Coating 95% | |
| Water Sensitivity ** | Retained Tensile Strength (AASHTO 283) | Min 80% | |

**Note:-**1. *Aggregate may satisfy requirements of either of these two tests.

2. **Water sensitivity test is only required if the minimum retained coating in the striping test is less than 95%

**PARTICULAR SPECIFICATIONS (CONTD.....)**

5.5.4 **FINE AGGREGATES**: Fine aggregates shall be the fraction passing IS 2.36 mm sieve and retained on 0.75mm sieve, consisting of 100% crushed manufacture sand resulting from crushing operations. It shall be non-plastic, clean, hard, durable of cubical shape It shall be free from soft pieces, sulphates, chlorides and other deleterious materials (including products of decomposition), which may be liable to break down during drying or subsequently when exposed to weather. The sand equivalent (IS-2720, Part-37) value for the fine aggregate shall not be less than 50.

5.5.5 **MINERAL FILLER**: OPC grade 43/ PPC of equivalent grade (as per specifications given here in before for concreting (general) shall be used as fillers.

5.5.6 **DESIGN CRITERIA**: Semi Dense Asphaltic concrete mix should be properly designed so as to satisfy certain criteria needed to assure adequate stability and durability. The mix as designed and laid should satisfy the requirements as given under:-

| **CRITERIA** | **SPECIFIED VALUE** (Wearing Course) |
|---|---|
| Number of compaction blows on each end of Marshall specimen | 75 |
| Marshall stability in, kN at 60 degree C, min | 14 |
| Flow value (mm) | 2-4 |
| Percentage void in mix (to prevent bleeding) | 3-5 |
| Percentage void in mineral aggregates filled with bitumen. | 65-75 |
| Binder content % by weight of total mix. | 5.5% |
| Percent voids in the mineral aggregate (VMA) | As per Table-500-12 of MORT&H |
| Retained tensile strength % (AASHTO T 283) | Min 80% |

**Note**: - Tensile Strength Ratio (TSR) shall be determined in accordance with AASHTO T 283. If TSR of mix conditioned with one freeze and thaw cycle is less than 80%, 2% hydrated lime or liquid anti-striping agent shall be used in the mix to satisfy this criteria, cost of which shall deemed to be included in quoted rate.

5.5.7 **IMMERSION COMPRESSION TEST**: -Compression test to measure the loss of Marshall Stability due to effect of water on the mix shall be conducted and if the index of retained stability is less than 75, the mix shall be rejected or the aggregates shall be processed to increase the index to minimum 75. The immersion test shall be conducted on Marshall test specimens of design job mix after storing in 1% sodium chloride solution v/v distilled water for 24 hours at $60^0$C and then tested for Marshall Stability values. The percentage loss or retention of stability values after immersion in water with reference to the values obtained before immersion is expressed in terms of percentage in the original values.

5.5.8 **JOB MIX FORMULA**:-

a) The blending of various types and sizes of aggregate shall be done on the basis of the requirements as indicated and optimum bitumen content within the specified limits.

b) The tenderer shall submit their job-mix formula to the Garrison Engineer before 30 days of start of SDAC work duly tested from CRRI, New Delhi/ IITs/ NITs/Govt Engg College/Highway Research Centre /SEMT Wing, CME Pune, National test house giving the details as mentioned in the preceding clauses for approval of the Job-mix formula for actual use.

**PARTICULAR SPECIFICATIONS (CONTD.....)**

c) The approved Job-mix formula shall remain effective unless and until the same is modified and re-approved by the Garrison Engineer. More than one Job-mix formula is permissible based upon the properties of actual materials available during construction of the project and also due to change in sources of material, if any.

d) For any change in source of supply of aggregate, fresh mix design shall be got done and testing charges shall be borne by the contractor.

e) SDAC shall be designed to satisfy the criteria as specified in clause here-in –before.

5.5.9 **PERMISSIBLE TOLERANCES**: The permissible variations of various ingredients shall be within the following limits:-

| Description | Permissible variation | |
|---|---|---|
| | Base/binder course | Wearing course |
| Aggregate passing 19mm sieve or larger | +8% | +7% |
| Aggregate passing 13.2mm, 9.5mm | +7% | ±6% |
| Aggregate passing 4.75mm | ±6% | ± 5% |
| Aggregate passing 2.36mm, 1.18mm, 0.6mm | ±5% | ± 4% |
| Aggregate passing 0.3mm, 0.15mm | ±4% | ± 3% |
| Aggregate passing 0.075 | ±2% | ± 1.5% |
| Binder content | ±0.3% | ± 0.3% |

5.5.10 **MIXING AND TRANSPORTATION OF MIX**:

a) The temperature of binder at the time of mixing shall be in the range of 150 degrees Celsius to 165 degrees Celsius and of the aggregates in the range of 150 degrees Celsius to 170 degrees Celsius, provided that at no time, the difference in temperature between the aggregates and binder shall exceed 14 degrees Celsius. The mixing shall be thorough to ensure that a homogeneous mixture is obtained in which all particles of the mineral aggregates are coated uniformly and temperature of mix shall not exceed 165 degrees Celsius.

b) Hot Mix Plant shall be used for mixing of aggregate and the binder. The binder shall be heated to the specified temperature. The aggregate shall be suitably warmed or heated before loading into hot mix plant. The correct quantity of each size of aggregate shall be fed into mixer with specified quantity of binder.

c) The mix shall be transported from the mixing plant to the point of use in suitable tipper vehicles specified here-in-after. The vehicles employed for transport shall be clean and be covered using suitable covers in transit to ensure that the temperature of mix does not fall below 125 degrees Celsius at the time of laying**.**

5.5.11 **SPREADING OF THE MIX**: Spreading of SDAC shall be done by means of self-propelled mechanical paver. DAC shall not be laid during rainy weather or when the surface is damp or wet. The transverse joints or construction joints shall be placed in the vertical plane after cutting back to the original thickness of previously laid mix. The vertical cut faces shall be painted with bitumen (VG-30 Grade) prior laying of fresh mix.

**PARTICULAR SPECIFICATIONS (CONTD.....)**

5.5.12 **ROLLING AND COMPACTION**:

a) The rolling and compaction process for Asphaltic Concrete (SDAC) shall be in four stages as described here in after. The rolling shall be done with 8 to 10 tonne Roller used on static mode.

b) Rolling operations shall be conducted when the mix is neither too hot nor too cold, so that shoving or hair cracks may be eliminated. Rolling shall be continued till the density is achieved at least 98% of that of the Laboratory Marshall Specimen and all roller marks are eliminated.

c) Rolling operations shall be completed in all respects before the temperature of the mix falls below 100 degrees Celsius.

d) The wearing surface shall be cold rolled with pneumatic tyred roller for a minimum of 100 passes in a period of 3 month or opening to traffic whichever is earlier and as directed.

5.5.13 **PAVEMENTS EDGES**: Pavement shall be laid to correct width and alignment. To achieve straight and vertical edges, the contractor shall cut back the edges to correct width and alignment by removing extra mix spread. Nothing extra shall be paid on this account.

5.5.14 **TEST FOR BINDER CONTENT**: The binder content of the mix shall be tested periodically and as directed by the GE using centrifugal extraction method or such method approved by the GE. The test shall generally be carried out in accordance with established practices and as directed by the GE. Whenever, there is a deviation in the resultant binder content from approved job-mix-formula values, it shall be reported to the GE immediately and necessary correction made at the job-mix plant as directed.

5.5.15 **RATE**: Unit rate quoted for SDAC shall include for cost of all material, plant/tools/machineries and labour required in all operations described here-in-before. Rate shall include cost of all materials, plant, machinery and labour required in all the operations described above including all cartages and lifts involved.

5.6. **TACK COAT**: Tack coat work shall be carried out as specified in Sch"A" and as per para 20.A.24 including sub para 20.A.24.1, 20.A.24.2 and 20.A.24.3 of MES Schedule Part-I.

5.7. **BITUMINOUS MACADAM**:-

5.7.1 Refer Para 20.B.2.1, 20.B.2.5.1, 20.B.2.5.2, 20.B.2.5.3, 20.B.2.5.4 (b) and 20.B.3.1 to 20.B.3.13 of MES Schedule Part I.

5.7.2 The Bituminous Macadam Shall be properly got designed in a Laboratory as approved by the GE.

5.7.3 The proposed design mix shall be finalized and got approved by the GE before starting the work.

5.7.4 Laboratory Charges, cost of materials, its cartage and other incidental charges shall be borne by the contractor.

5.8 **THERMOPLASTIC ROAD MARKING**

5.8.1 Thermoplastic road marking paint shall be STP Ltd (Shalimark Brand), Asian Paints (Apcomark Brand), Kataline (Trakmark WR Brand).

5.8.2 The application of thermoplastic road marking shall be strictly as per Manufacturers Instructions.

**PARTICULAR SPECIFICATION (CONTD…./)**

5.8.3 Preparation of Surface: - The Surface shall be thoroughly cleaned and dusted. All dirt, scales, oil and grease shall be thoroughly removed before painting is started. The prepared surface shall be inspected and approved by the Engineer-in-Charge before painting is commenced. The weather shall not be foggy/rainy, dusty or windy. The suitability of the weather will be determined by the Engineer-in-Charge& GE.

6. **BRICK EDGING**

6.1 **Bricks**:-Bricks: Refer to Paras 5.6.1 to 5.6.9 of MES Schedule Part. I. The bricks shall have minimum compressive strength of 75 Kg/Sq cm and shall be of subclass 'B', old size bricks as specified in Para 5.6.2 and 5.6.3. of MES Schedule Part-I. Sampling and testing of bricks shall be carried out by the contractor as per IS-5454 and IS-3495 (Part -I to IV) as directed by GE without any extra cost to department. The size of bricks shall as per MES Schedule Part-I i.e. 23 x 11.3 x 7.5/25 x 12.5 x 7.50 cm as per availability. However the permissible tolerance in size shall be + 8% and centerline dimension shall be maintained.

6.2 Workmanship: - Refer Para 5.54 of MES Schedule Part-I.

7. **CONCRETE WORK**

7.1 **MATERIALS**

7.1.1 **CEMENT**

7.1.1.1 **TYPE OF CEMENT:** Cement shall be Ordinary Portland Cement Grade 43 conforming to IS 8112-1989 (Latest revision) or Portland Pozzollana Cement, conforming to IS 1489-1991 and shall not be more than 3 months old on the date it is received / brought at site work. For this purpose the date of manufacture printed on the cement bag and as mentioned in the purchase voucher shall be considered for calculating the period of 3 months.

7.1.1.2 Based on the CPM chart finalized, monthly requirement of cement and the proposed date wise Schedule of deliveries for bringing the cement at site shall also be worked out by the contractor and the same shall be submitted to the GE for information and approval, cement shall not be allowed to be stocked for more than two months requirement because of its deterioration with passage of time. Cement more than 6 months old shall be retested for checking its suitability for the use in works.

7.1.1.3 PROCUREMENT: Cement (OPC & PPC) supplied by the contractor shall be procured from following approved cement manufacturers / Main Producers:-"

| Ser No | Company Name | Brand | Type of Cement |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| **REGIONAL LEVEL** | | | |
| (i) | M/s Star Cements Meghalaya (Only for CE Shillong Zone & CE(AF) Shillong Zone area) | - | OPC & PPC |
| (ii) | M/s Meghalaya Cements Ltd (Only for Eastern Region) | TOP CEM | OPC & PPC |
| **PAN INDIA LEVEL** | | | |
| (i) | The Associated Cement Companies Ltd. | ACC | All |
| (ii) | M/s Ultra Tech Cement Ltd. | ULTRA TECH | All |
| (iii) | The India cement | - | All |
| (iv) | M/s Dalmia Cement (Bharat) Ltd | DALMIA KONARK | All |
| (v) | Century cements | CENTURY | All |

**PARTICULAR SPECIFICATION (CONTD….)**

| Ser No | Company Name | Brand | Type of Cement |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| (vi) | Saurashtra Cement | SAURASHTRA | All |
| (vii) | The Ramcocements Ltd | RAMCO | All |
| (viii) | Mangalam cement Ltd | MANGALAM | All |
| (ix) | Birla Corporation Ltd. | BIRLA | All |
| (x) | Orient cement | ORIENT | All |
| (xi) | Nuvoco Vistas Corporation Ltd | NUVOCO | All |
| (xii) | Shree cement | SHREE | All |
| (xiii) | J K Cement | J K | All |
| (xiv) | J K Lakshmi Cement Ltd. | J K LAKSHMI | All |
| (xv) | JaypeeRewa Cement | JAYPEE | All |
| (xvi) | Ambuja Cement Ltd. | AMBUJA | All |
| (xvii) | M/s Shree Guru Kripa Cement (Pvt) Ltd. | SARTAJ | OPC 43 Gr & PPC |
| (xviii) | M/s Parasakti Cements Ltd. | PARASAKTI | OPC 43 Gr & PPC |
| (xix) | M/S My Home Industries Ltd | MAHA CEMENT | OPC 43 Gr, PPC & PSC |
| (xx) | M/s Chettinad Cement Corporation Ltd | CHETTINAD CEMENT | OPC 43 Gr & PPC |
| (xxi) | M/s Sanghi Industries Ltd | SANGHI | OPC 53 Gr & PPC |
| (xxii) | M/s Wonder Cement Ltd | WONDER Cemenrt | OPC 43, OPC 53 & PPC |
| (xxiii) | M/s Kesoram Industries Ltd | BIRLA SHAKTI | OPC 43 Gr, OPC 53 Gr & PPC |
| (xxiv) | M/s JSW Cement Ltd, A.P | "JSW OPC 53" & "JSW OPC 43" | OPC 43 & OPC 53 |

**Note :-**

1. **Any other manufacturer(s) approved by E-in-C"s branch, New Delhi but left out in the above list shall also be deemed to be included in the list.**
2. **Any other manufacturer(s) approved by E-in-C"s branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution shall also be deemed to be included in the list.**
3. **If any of the above manufacturers is banned/ removed by E-in-C"s branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution, the same shall be deemed to be deleted from the list.**

7.1.1.3.1 The following conditions shall have to be satisfied before PPC can be allowed to be used in the work:-

(a) PPC should meet the strength criteria as per IS-8112-1989.

(b) In cold climate regions where temperature is lower than 15$_o$ C only OPC shall be used.

(c) Mixing of OPC & PPC shall not be allowed in a work. However with the strict control of ground executives different buildings can have different types of cement.

(e) While procuring PPC, the following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE for approval to procure PPC.

(i) The quality of fly ash is strictly as per IS-1489 (Part-I) – 2002

(ii) Fly ash is inter-ground with clinker not mixed with clinker.

(iii) Dry fly ash is transported in closed containers and stored in silos. Only pneumatic pumping should be used.

(iv) The fly ash received from thermal power plants using high temperature combustion above 1000$_o$C should be used.

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7.1.1.3.2 The particulars of the manufacturer/main producer of cement along with the date of manufacture shall be submitted by the contractor for every lot of cement separately. The documents in support of the purchases of cement shall be verified by the site staff and GE.

7.1.1.4. **TESTING OF CEMENT:** The manufacturer/ main producer shall carry out inspections and testing of cement in accordance with the relevant BIS provisions. The Contractor shall submit the Manufacturers Test Certificate in original along with the Test Sheet giving the result of each physical test as applicable and the chemical composition of the cement or authenticated copy thereof, duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in the Cement supply and Acceptance Register, after due verification. The GE shall also organize independent testing of random samples of cement drawn from each lot from the NATIONAL TEST HOUSE, SEMT CME, GOVERNMENT APPROVED LABORATORIES, ZONAL LABORATORIES as per IS 3535-1986 (Method of sampling hydraulic cement), IS-4031 (Method of physical test for hydraulic cements) and IS-4032-1985 (Method of chemical analysis of hydraulic cement). In order to undertake, departmental testing, requisite facilities alongwith materials, conveyance etc. shall be organised by the contractor without any extra cost to the Govt. The cost of testing, conveyance etc shall be borne by the contractor.

7.1.1.5. **STORAGE:** Cement shall be stored over dry platform at least 20 cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of store rooms, the stack shall be at least 20cm away from floors and walls. Inspection of storage shall be carried out by Engineer-in-Charge or J.E (Civil) specifically detailed by him in writing, once a day. It shall be ensured that tested and untested cement are segregated and stored separately with distinct identification.

7.1.1.6 **DOCUMENTATION:** The Contractor shall submit original vouchers from the main producer/manufacturers for the total quantity of cement supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the GE along with the relevant documents before acceptance. The original vouchers and the Test Certificates shall be defaced by the Engineer-in-Charge and kept on record in the Office of GE duly authenticated and with cross reference to the Control Number recorded in the cement Supply & Acceptance Register. The Cement Supply & Acceptance Register will be signed by J.E. (Civil), Engineer-in-Charge, GE and the Contractor. The Accepting Officer may order Board of Officers for random check of cement and verification of connected documents. The entire quantity of all cement shall also be suitably recorded in the Measurement Book for record purposes before incorporation in the work and shall be signed by the Engineer-in-Charge and the Contractor.

7.1.1.6.1 Identification of cement bags by the manufacturer shall be as per relevant BIS codes.

7.1.1.6.2 The entire lot of cement brought by contractor to the site shall be removed by contractor under his own arrangements and at his own cost, whenever the test results are not found within the prescribed acceptable limits. Decision of GE in this respect shall be final and binding.

7.1.1.6.3 The cost of all tests including cost of transportation shall be borne by contractor irrespective of test results.

7.1.1.6.4 The proper accommodation for storage, preservation and maintenance of cement shall be provided by the contractor at his own cost till consumption of cement in the work at the place(s) decided by GE. All arrangement for proper storage and safety of cement as indicated here-in-before shall be made by contractor at his own cost. The cement storage accommodation shall be provided with two locks at each door. The key of one lock at each door shall remain with the Engineer-in-Charge or his representative and that of other lock shall remain with the contractor's authorised agent at site of works so that cement is removed from the go-down/storage accommodation according to daily requirement with the knowledge of both the parties.

7.1.1.6.5 Schedule of supply of cement shall be worked out by contractor in consultation with GE so as to match with the progress contemplated in CPM Chart prepared for this Project.

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7.1.1.7 **WEIGHING AND PAYMENT FOR CEMENT**

7.1.1.7.1 Random samples of 5 bags per 100 bags shall be got weighed by Engineer-in-Charge in presence of representative of contractor from each consignment to arrive at average weight per bag for calculating total quantity of cement brought at site for incorporation in the work. All arrangement for weighing of cement bags including labouretc shall be provided by the contractor at his own cost.

7.1.1.7.2 The quantity of cement arrived at as aforesaid shall be considered for payment as applicable in accordance with IAFW-2249, General Conditions of Contracts.

7.1.1.7.3 Payment for cement shall only be allowed after production of original purchase vouchers, manufacturer's test certificates for each consignment of cement brought by contractor and after the results of random testing carried out by GE are found satisfactory as specified herein-in-before.

7.1.2 **AGGREGATES**

7.1.2.1 **COARSE AGGREGATE (STONE)**: Aggregate for all concrete work shall be crushed stone aggregate conforming to IS-383 and para 4.4 of MES Schedule Part I, 2009. The aggregate also shall be in conformity with the samples kept in GE"s office.

7.1.2.2 **FINE AGGREGATE (COARSE SAND)**: Fine aggregate shall be sand conforming to the requirements to IS-383 and para 4.4 of MES Schedule Part I Sand shall be washed in clean water if the deleterious materials exceed the permissible limit. Use of sand confirming to grading Zone-IV of IS-383 shall not be allowed in RCC/PCC work.

7.1.2.3 Grading of coarse aggregate unless otherwise specified shall be as follows:-

| SL.NO | DESCRIPTION | | GRADING |
|---|---|---|---|
| (a) | For Reinforced/Structural Element of depth/thickness 50mm or under | : | 12.5mm graded |
| (b) | For Reinforced/Structural Element of depth/thickness over 50mm | : | 20mm graded |
| (c) | For Plain Cement Concrete thickness exc. 25mm but not exc. 75mm | : | 20mm graded |
| (d) | For Plain Cement Concrete thickness 75mm and over | : | 40mm graded |
| (e) | Lime Concrete | : | 40mm graded |

7.1.2.3.1 Grading of coarse aggregate for plain and reinforced concrete i.e. stone aggregate shall be as per para 4.4.7.1 of MES Schedule part-I and grading of fine aggregate sand shall be as per para 4.4.7.2 of MES Schedule Part-I

7.1.2.4 **AGGREGATES FROM NATURAL SOURCES**: Aggregates obtained from natural sources shall not be contained any type of harmful materials and shall be cleaned in all respect.

7.1.3. **WATER**: Refer para 4.9 & 4.9.2 of MES Schedule Part I. Water shall conform to clause 5.4 IS-456-2000 in case water is drawn by the contractor from his own sources as mentioned in Special Condition Clause No 10 of these tender documents.

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7.1.4 **PLASTICISER**

7.1.4.1 Plasticiser shall be used in Design Mix Concrete as per the Manufacturer's instruction without any Extra Cost to achieve the desired Results Strength and Workability. However Plasticiser should confirm to IS-9103.

**8. DEMOLITION/DISMANTLING/MAKING GOOD**

8.1 Refer paras of section 21 of MES Schedule Part-I. All demolished/taking down materials (except those were re-used and not accounted in Schedule of Credit) shall be returned to MES store without any extra cost to the Government.

8.2 Please note that if any existing surface is damaged during taking down/execution of work that surface shall be made good as per existing one without any extra cost to the Govt. The rate for making good is deemed to be inclusive in the quoted rates of Schedule 'A'/BOQ.

9. **PROTECTION DURING EXECUTION**

9.1. During execution of work adequate preparation and care shall be given so that the existing structures and other allied items are not damaged. If anything happens in this regards, the same has to be made good by the contractor at his own expense.

10. **REMOVAL/SITE CLEARANCE**: Old demolished materials considered under Schedule of credit shall be the property of the contractor and will be removed by the contractor from the site. Debris and other materials, if any, shall be disposed off and sites to be cleared to the entire satisfaction of the GE.

11 **PRODUCTION OF PURCHASE VOUCHERS:**

The contractor shall submit original purchase vouchers, invoice from original manufactures only in respect of purchase of items for which makes have been specified and incorporated in the contract to the Engineer-in-Charge who shall be responsible to verify and deface the same before according his permission to incorporate such material in the work. These original purchase vouchers / invoices shall be released to the contractor duly defaced by Engineer-in-Charge after technical check of final bill by Garrison Engineer. In exceptional cases where the contractor demands for return of the original vouchers / invoices for submission to income tax / sales tax department prior to technical check of final bill by the Garrison Engineer then Engineer-in-Charge, if satisfied about the genuineness of the requirement may return the original vouchers / invoices duly defaced to the contractor. However, photo copy of each such vouchers / invoice shall be retained by Engineer-in-Charge.

11.1 **RECORD OF MATERIALS AND PURCHASE VOUCHERS:-**

11.1.1 The quantity of materials such as cement, steel, paints, water proofing compound, chemicals for anti- termite treatment and the like, as directed by the Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work. Voucher/invoices in respect of cement, steel and major E/M equipments like transformers, DG sets, pumps, motors, AC & lift equipments shall invariably be submitted by the contractor. However contractor shall produce vouchers/invoices from the manufactures and or their authorized agents for the full quantity of all the materials as applicable as a pre-requisite before submitting claims for payments for advances on account for the "Material laying at site" in accordance with condition 64 of IAFW-2249, General Condition of Contracts.

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11.1.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in measurement book shall be suitably marked for identification.

11.1.3 Contractor shall also produce vouchers/invoices from the manufacturers and/or their authorized agents for the full quantity of the specific materials/additional items, as applicable as a pre-requisite before submitting claims for payment for advances on account of the work done and/or materials collected in accordance with Condition 64 of IAFW-2249, General Conditions of the Contracts.

The list of such additional items for which accepting officer may consider necessary to submit purchase vouchers for verification against the subject contract are as under:-

(a) Thermoplastic paints/Road marking strips

11.1.4 The contractor shall ensure that the materials are brought to site, in original sealed container/packing, bearing manufacturer's marking except in the case of the requirement of materials (s) being less than smallest packing.

11.1.5 Bitumen shall be purchased directly from main producers only as specified here-in-before. These vouchers / invoices shall be defaced by the GE/Engineer-in-Charge, indicating reference to the contract number, under his dated signature, and CTC thereof shall be kept on record so as to avoid their being used again.

11.1.6 The vouchers/invoices will clearly indicate the contract number and the IS No, specific alternative to which the material conforms in case of various alternative in IS/BIS.

12. **STRUCTURAL STEEL**

(a) Definition of structural steel as given in clause 10.4 of SSR Part-I shall be applicable. Standard quality steel of grade E-250(Fe-410W quality A) conforming to IS-2062 for all types of steel structures including those subject to dynamic loading shall be used.

(b) Ordinary quality structural steel wherever mentioned shall be conforming to IS-1977 of grade E-165(Fe-290) and shall be used for steel doors, collapsible gate, window guard bars or grills, steel gates, hand railings, fencing posts etc.

(c) Unless otherwise mentioned, galvanized steel sheets (plain or corrugated) where applicable shall be conforming to IS-277. Grade of zinc coating to be used shall be the one which is given in Clause 10.29.1 of SSR Part-I.

12.1 **SOURCE OF PROCUREMENT**

12.1.1 **STRUCTURAL STEEL:-** Structural steel shall be procured by the contractor from approved primary manufacturers / main producers of steel like **Rashtriyalspat Nigam Limited (RINL), Visakhapatnam / TATA Iron and Steel Company (TISCO or TATA Steel), Mumbai / Steel authority of India Ltd (SAIL), Delhi/ M/S Jindal Steel & Power Ltd, Gurgaon.**

**Note:-**

1. **Any other manufacturer(s) approved by E-in-C"s branch, New Delhi but left out in the above list shall also be deemed to be included in the list.**
2. **Any other manufacturer(s) approved by E-in-C"s branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution shall also be deemed to be included in the list.**
3. **If any of the above manufacturers is banned/ removed by E-in-C"s branch, New Delhi during the pre-tender and post tender stage/after acceptance/during execution, the same shall be deemed to be deleted from the list.**

**PARTICULAR SPECIFICATION (CONTD….)**

12.2 **TESTING OF STEEL**

12.2.1 The manufacturer is to carry out inspection and testing of steel in accordance with the relevant BIS provisions. The contractor shall submit the manufacturer's test certificate in original along with the Test Sheet giving the result of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in Steel supply & Acceptance Register (as per proforma given herein-after) after due verification and send a certified true copy of test sheet to GE(I)/GE/AGE(I) for his records. The GE/ CWE shall also organize independent testing of random samples of steel drawn from various lots from National Test House, SEMT Wing CME, Regional Research Labs and NABL approved Labs etc as per the recommended minimum frequency shown in table as per given here-in-after. Samples from each lot should be tested for quality and elongation. The elongation shall be as admissible as per IS. Cost of samples, transportation and testing shall be borne by the contractor. The record of such checks would be maintained in the steel test register.

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12.2.2 **Independent testing**

12.2.2.1 The GE/CWE shall also organize independent testing of random samples of steel drawn from various lots from National Test House, SEMT Wing CME Pune, Regional Research Laboratories, Govt approved laboratories, Zonal laboratories as per the recommended minimum frequency, in addition samples from each lot shall be tested for quality and elongation as per the parameters given in STEEL SUPPLY & ACCEPTANCE FORM. The elongation shall be as admissible as per IS.

13 **BRICK WORK**

13.1 Cement: Shall be as specified for concrete work.

13.2 Sand for mortar: refer to para 5.4, 5.4.1 of MES Schedule Part-I.

13.3 Bricks: Locally available best quality bricks shall be used. Refer to paras 5.6.1 to 5.6.9 of MES Schedule Part-I. The bricks shall have minimum compressive strength of 75 Kg/Sq cm and shall be of subclass 'B', old size bricks as specified in para 5.6.2 ,5.6.2.1 and 5.6.3. of MES Schedule Part-I. Sampling and testing of bricks shall be carried out by the contractor as per IS-5454 and IS-3495 (Part -I to IV) as directed by GE without any extra cost to department. The size of bricks shall as per MES Schedule Part-I i.e. 23 x 11.3 x 7.5 cm. However the permissible tolerance in size shall be + 8% and centerline dimension shall be maintained.

13.4 All brick work shall be built in English bond except in half brick walls which shall be in stretcher bond.

13.5 **HALF BRICK WALLS - REINFORCED**:

13.5.1 Half brick thick wall of height more than 900 mm shall be reinforced with 2 Nos. 8 mm dia TMT Bars horizontally at every fourth course starting from floor level and anchored in walls/ columns at junction, unless otherwise specified.

13.5.2 The foundation for Half Brick thick walls shall be approved as shown in Foundation Plan or. Where foundations for Half Brick thick walls is/ are not shown in drawings/ sketch at site, the same shall be constructed sub-base level. Half Brick thick walls in First Floor and onwards shall be constructed from RCC Floor slab.

13.6 **WATER**

Refer to paras 5.18 of MES Schedule part-I.

13.7 **BRICK WORK**- Mix of mortar for brick work unless otherwise specified here-in-after, shall be as under: -

(i) Brick work in half brick wall, isolated pillars, parapet walls, steps shall be built in cement mortar (1:4).

**PARTICULAR SPECIFICATION (CONTD….)**

(ii) All other brick work in various situations unless otherwise specified shall be built in cement mortar (1:6).

13.8 **FILLING OF HOLES**:-Any hole left in the brick work for the purpose of Centering/Scaffolding shall be filled up with Cement Concrete 1:3:6 (using 20mm graded crushed Stone aggregates) and mixed with WPC as per Manufacturer's instructions.

14. **PAINTING**

14.1 **PAINTS AND ALLIED MATERIALS**

(i) Refer para 17.2 of MES Schedule Part-I.

(ii) Paint shall be of first quality and manufactured by Standard firms are ASIAN/ BERGER/ ICI/ JHONSON & NICHOLSON/ NEROLAC/ SHALIMAR

14.2 **PAINTING WORK GENERALLY**

Refer para 17.3 and subsequent thereof MES Schedule Part-I.

14.3 **PAINTING TO STEEL AND IRON WORK**

(i) Refer para 17.8 of MES Schedule Part-I.

(ii) Filler coat is not required to be applied. Where painting is indicated in Schedule of finishes, it shall be done in three coats viz primer, under coat and finishing coat. Primer for steel surfaces shall be red oxide zinc chrome and under coat and finishing coat shall be of synthetic enamel paint.

(iii) All exposed surfaces of iron and steel other than galvanised surfaces and reinforcement shall be painted with two coats of synthetic enamel paint over a coat of primer. Primer for steel work shall be red oxide zinc chrome primer. Steel brackets of cisterns/WHB shall be given two coats of aluminium paint over a coat of primer. However, steel members of roof, hidden by ceiling shall be painted with two coats of red oxide primer.

15. **EXTERIOR EMULSIONPAINT (WEATHER PROOF PAINT):-**

15.1 External plastered surfaces of walls/RCC/cill/jamb/RCC facia of chajja/Roof/RCC wall/parapet/soffit of slab shall be applied exterior emulsion paint as mentioned in Schedule of Works/ BOQ and as per clause No 17.17 of SSR Part-I 2009.

15.2 Newly plastered surfaces shall be scrapped to remove dust, loose or flaking materials, grease & dirt to ensure a smooth and clean surface. Priming shall be done as per specified in Schedule of Works/ BOQ and as approved by GE. Paint shall be thinned down as per manufacturer's instructions. Application shall be done using brush or roller. The recoating time between two coats shall be around 1 to 6 hours.

16 **PLASTERING AND POINTING**

16.1 **CEMENT**- Cement shall be as specified here-in-before.

16.2 **SAND** – Refer para 14.5, 14.5.1 and 14.5.2 of MES Sch Part-I.

17 **WORKMANSHIPS**

17.1 **GENERALLY**

17.1.1 Refer to Clause No.14.16 of MES Schedule Part-I (as applicable) and as directed by the Engineer- in-Charge.

17.1.2 The thickness of plaster shall be as specified in Schedule 'A' and shall be finished thickness exclusive of dubbing out. Plastering and dubbing out shall however, be done in one operation.

17.1.3 The external and internal finishes (Plastering and pointing) shall be returned in the Jambs of the openings up to frames and taken around cup boards and other similar projections.

17.1.4 External plaster shall be finished fair and even. Internal plaster shall be finished even and smooth without using extra cement.

**PARTICULAR SPECIFICATION (CONTD….)**

17.1.5 All corner, angle, junctions and edges shall be truly vertical or horizontal as the case may be and shall be carefully finished. Corners around jambs of opening and junctions of wall plaster shall be rounded to minimum radius of 5mm. 12mm wide groove to the entire thickness of plaster shall be provided at the junction of walls and RCC beams. 10mm wide and 10mm deep shall groove shall also be provided at junctions of walls and RCC columns or any other dissimilar materials.

17.2 **SCAFFOLDING** – Refer Clause No.14.12 of MES Schedule Part-I and all as directed by Engineer-in-Charge.

17.3 **PREPARATION OF MORTAR FOR PLASTERING** – Refer Clause No.14.13 of MES Schedule Part-I as applicable and all as directed by Engineer-in-Charge.

17.4 **CURING** – Refer Clause No.14.24, 14.24.1 of MES Schedule Part-I and as directed by Engineer-in-Charge.

17.5 **WHITE WASHING/COLOUR WASHING AND CEMENT BASE PAINT**

17.5.1 **LIME**

17.5.1.1 Refer para 15.2 of MES Schedule part I.

17.6 **BLUE VITRIOL**

17.6.1 Refer paras 15.4 of MES Schedule Part I.

17.7 **WHITE WASHING**

17.7.1 Refer para 15.12 and subsequent para thereof MES Schedule Part I.

17.7.2 Where white washing is indicated in Schedule of finishes, 3 coats of white wash shall be applied. The gable walls above ceiling shall be finished with three coats of white wash including preparation of plastered surfaces.

18.4 **CEMENT BASE PAINT**

18.4.1 Refer paras 15.7 of MES Schedule Part I.

18.4.2 Where cement base paint is mentioned in schedule of finishes two coats of cement base paints shall be applied. Cement base paint shall be of superior quality manufactured by standard firms are as ACROCEM/ DUROCEM/ SUPER SNOWCEM/ Berger (XTRCE M-76 Superior Cement paint) / Asian Paint (Gutucam)

19 **DISTEMPEROIL EMULSION:-**

19.1 Distemper-oil Emulsion shall be 1st quality and conform to IS-428-1969 and as specified in clause No. 15.6. These shall be carried out all as specified in clause 15.14 MES SSR Part-I, 2009 and as directed by the EIC. Distemper of required colour and shade shall be obtained ready mixed conforming to IS-428-1969. The contractor shall approach GE in writing through EIC regarding brand & shades of oil bound distemper, which he wants to use in the work & after obtaining prior approval from GE in writing. Contractor shall be allowed to execute the work in full, if any portion of work is executed without complying the above requirement, contractor shall redone the work as ordered by GE without any extra cost of the department .

20. **WET MIX MACADAM:-**

20.1 Refer paras 20.B.2.5 and 20.B.5 of MES Schedule of Rates Part-I, 2009.

## PARTICULAR SPECIFICATIONS (CONTD)

## LIST OF PRODUCTS FOR WHICH SPECIFIC MAKES ARE TO BE INCORPORATED IN WORKS

### 12.GENERAL

Materials to be incorporated in the Work shall be any one of the following MAKE as approved by the GE in writing before incorporated in the Work. These shall strictly comply with latest appropriate Indian Standard Specification or where IS Specification is not available these shall comply with the latest BS Specification.

| SL NO | DESCRIPTION OF ITEMS | MAKE/BRAND |
|---|---|---|
| 1 | 2 | 3 |
| 1 | Cat eyes/Studs(Metal/Plastic) Rumble Strip/Speed Breakers, Delinator/Road Barriers & Road Signage | 3M / Greenlite / Dark Eye / Autoamat |
| 2 | Solar Cat Eye | 3M / Tata BP Solar |
| 3 | Thermo Plastic Road Marking Paints/Retro-Reflective paint | Asian Apcomark / Godavari Paints Pvt Ltd / STP / Sunrise / Shalimar apints Ltd / Faith / Berger paints Ltd / Automark / Jensong& Nicholson Paints Ltd / Sunny / Jotun India Pvt Ltd. |

**Note:**

1. Makes specified in Sch 'A' with cat Part Nos. shall only be provided among the other makes are mentioned in Sch 'A'. If the items mentioned cat Part in Schedule ‘A’ is not available then the equivalent one shall be chosen based on documentary evidence from the other makes mentioned in Schedule ‘A’ against that item.
2. Makes specified in Particulars Specifications shall be provided only when no makes are specified in Sch‘A’.
3. Makes specified in this Appendix shall be provided only when no makes are mentioned in Sch‘A’ and Particulars Specifications.
4. Contractor will supply the material only after obtaining the written approval from GE regarding use of specific make of items and will supply the material at appropriate time as and when required by the GE.
5. To ensure genuineness of material the contractor will order only on manufacture or 111 authorized dealer and the Engineer-in-Charge will receive material after verifying the delivery challan of the supplier and after satisfying the genuineness of material.
6. **All other materials:-** Make and sample to be approved by GE

**SIGNATURE OF THE CONTRACTOR** **DCWE (CONTRACTS)**

**DATED:** **FOR ACCEPTING OFFICER**

**PARTICULAR SPECIFICATION-II (CONTD)**

**APPENDIX-`A'**

**CEMENT SUPPLY/ACCEPTANCE FORM CEMENT ACCEPTANCE REGISTER**

1. CA No & Name of work:
2. Control No
3. Name of Manufacuterer/Brand Name/ Gde of Cement (A) Manufacture _____ (b) Brand _____ (c) Grade
4. Qty of cement & Lot No/Week No (in Bags) :Qty _____ (b) Lot No/Week No _____
5. Manufacturer's test Certificates No _____
6. Random Test Details (a) Physical test report from vide their letter No _____

   (Name of approval Lab/Engg College)

   (b) Chemical test report from _____ vide their letter No _____

   (Name of approval Lab/Engg College)
7. Details of Physical & Chemical properties:-

**SIGNATURE OF CONTRACTOR** **DCWE (CONTRACTS)**

**DATED** **FORACCEPTING OFFICER**

**PARTICULAR SPECIFICATION-II (CONTD)**

**APPENDIX-`A' (Contd...)**

## CEMENT SUPPLY/ACCEPTANCE FORM CEMENT ACCEPTANCE REGISTER

| | Physical Requirements (As per IS 4031)1 | | | | | | | | | | Chemical Requirements (As per IS 4032) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Compressive Strengths | | | | | | | | | | | | |
| | Specific Surface ($M^2$/Kg) | Soundness by Le chatellier Expansion Expansion (%) | Soundness by Auto Clave Expansion | Initial setting time (Minutes) | Final setting time (Minutes) | 03 Days (Mpa) | 07 Days (Mpa) | 28 Days (Mpa) | Temp during testing 0C | Standard consistency (%) | Lime saturation factor (Ratio) | Almina Lon Ratio (Ratio) | Insoluble Resdue (%) | Magnesia (%) | Sulphuric Anhydride (%) | Loss opn Ignition (%) | Alkales (%) | Chloride (%) |
| As per relevant IS | | | | | | | | | | | | | | | | | | |
| As per manufacturer's test certificate | | | | | | | | | | | | | | | | | | |
| As per random test certificate | | | | | | | | | | | | | | | | | | |

**SIGNATURE OF CONTRACTOR** **DCWE (CONTRACTS)**

**DATED** **FORACCEPTING OFFICER**

**PARTICULAR SPECIFICATION-II (CONTD)**
**STEEL SUPPLY & ACCEPTANCE REGISTER**

**APPENDIX-`B'**

1. CA No& Name of Work
2. Contract No
3. Name of Manufacturer's T.C No
4. Manufacturer
5. Random Test details (a) Physical test report from _____ vide their letter No _____
   (Name of approved Lab/Engg College)
   (b) Chemical test report from _____ vide letter No_____
   (Name of approved Lab/Engg College)
6. Types of steel, Dia & Qty (a) type : TMT/CRS (b) Dia_____ mm (c) Actual Wt _____ MT (d) Conversion Wt _____ MT

| | Chemical test | | | | | | Mechanical test | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Carbon % | Sulphur % | Phos phorous % | Manganese % | Silicon % | Corrosion Resistant element | Wt per metre | Stress (N/mm$^2$) 0.2% proof stress | Tensile strength (N/mm$^2$) | Elongation% | Bend Test | Rebend Test | Re mar ks |
| As per IS 1786 | | | | | | | | | | | | | |
| As per Manufacturer's test certificate | | | | | | | | | | | | | |
| As per independent test | | | | | | | | | | | | | |

Remarks with Signature
Accepted/Rejected

Contractor Junior Engineer Engineer-in-Charge Garrison Engineer

Remarks of BOO/Inspecting Officer/CWE

**SIGNATURE OF CONTRACTOR** **DCWE (CONTRACTS)**

**DATED** **FORACCEPTING OFFICER**

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'C'**

| Sl No | Materials | Test | | Method of testing | Frequency of Tests | Level of test | Rate per test Rs. Ps. | Remarks |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | | 4 | 5 | 6 | 7 | 8 |
| 1. | **Bricks** | (i) | Compressive strength Test | IS-3495 (Part-II) | As per IS-5454 as given under:<br>Lot size / Sample size / Permissible No of defective bricks.<br>1001 to 10000 / 5 / 0<br>10001 to 35000 / 10 / 0<br>35001 to 50000 / 15 / 1 | 'A' | 180.00 | Checks for visual and dimensional characteristics shall also be carried out. |
| | | (ii) | Water absorption | IS-3495 (Part-II) | | 'A' | 150.00 | As per IS-5454. |
| | | (iii) | Efflorescent | IS-3495 (Part-I) | | 'A' | 180.00 | Legend<br>A-Site lab |
| 2. | **Coarse aggregate** | (i) | Sieve analysis | IS-2386 (Part-I) | One test for every 15CuM of aggregates or part thereof brought to site. | 'A' | 120.00 | |
| | | (ii) | Flakiness Index | -do- | -do- | 'A' | 90.00 | |
| | | (iii) | Estimation of deleterious materials | -do- | One test for every 100 CuM of aggregates or part thereof | 'A' | 120.00 | |
| | | (iv) | Organic impurities | -do- | One test per source of supply | 'C' | 120.00 | |
| | | (v) | Moisture content | IS-2386 (Part-II) | Regularly as required | 'A' | 120.00 | |
| | | (vi) | Specific gravity | -do- | One test for each source of supply | 'B' | 120.00 | |
| 3. | **Fine aggregate** | (i) | Sieve analysis | IS-2386 (Part-I) | One test for every 15 Cum of FA or part when brought to site. | 'A' | 180.00 | |
| | | (ii) | Test for clay, silt and impurities | IS-2386 (Part-I) | -do- | 'A' | 90.00 | |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'C' (Contd…)**

| Sl No | Materials | Test | | Method of testing | Frequency of Tests | Level of test | Rate per test Rs. Ps. | Remarks |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | | 4 | 5 | 6 | 7 | 8 |
| | | (iii) | Specific gravity | IS-2386 (Part-I) | One test for each source of supply | 'B' | 180.00 | |
| | | (iv) | Moisture content | IS-2386 (Part-II) | Regularly as required subject to 2 tests/day when being used. | 'A' | 180.00 | |
| | | (v) | Test for organic impurities | IS-2386 (Part-II) | One test for each source of supply | 'C' | 180.00 | |
| 4. | **Structural Concrete (M-15 grade and above)** | (i) | Slumptest or compacting factor test or vee-bee time. | IS-1199 | The minimum frequency of sampling of concrete of each grade shall be as under :- | 'A' | 180.00 | Random sampling shall be carried to cover all mix units. |
| | | (ii) | Compressive strengths | IS-516 | Qty of concrete in the work (Cubic Meter)<br>Qty of concrete in the work (Cubic Meter) / No of Samples<br>1-5 / 1<br>6-15 / 2<br>16-30 / 3<br>31-50 / 4<br>51 and above / 4+1 for each Addl 50 Cum or part thereof. | 'A' | | 120.00 |
| 5. | **(a) PCC Block for walling (Hollow Block)** | (i) | Compressive strength | IS-2185 | 08 Blocks out of 14 | 'A' | 60.00 | Sample:- 14 blocks from consignment of every 5000 blocks or part thereof. |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'C' (Contd…)**

| Sl No | Materials | Test | | Method of testing | Frequency of Tests | Level of test | Rate per test Rs. Ps. | Remarks |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | | 4 | 5 | 6 | 7 | 8 |
| | | (ii) | Water absorption | -do- | 03 Blocks out of 14 | 'B' | 120.00 | |
| | | (iii) | Density | -do- | 03 Blocks out of 14 | 'B' | 90.00 | |
| | **(b) PCC solid Block for walling** | (i) | Compressive strength | IS-2185 | 12 Blocks out of 18 | 'A' | 60.00 | Sample :- 18 blocks from consignment of every 1000 blocks or part thereof. |
| | | (ii) | Water absorption | -do- | 03 Blocks out of 18 | 'B' | 120.00 | These Block to be checked for dimension and weight. |
| | | (iii) | Density | -do- | 03 Block out of 18 | 'B' | 120.00 | |
| 6. | **Cement Flooring Tiles/Ter-razzo Tiles** | (i) | Water absorption | IS-1237 (Appx'D') | 06 tiles out of 18 | 'B' | 180.00 | Sample of 18 tiles from each source of supply selected at random. |
| | | (ii) | Wet transverse strength | -do- (Appx'E') | -do- | 'B' | 144.00 | |
| | | (iii) | Resistance to wear | -do- (Appx'F') | -do- | 'C' | 540.00 | |
| 7. | **Burnt clay roofing tiles (hand made) as per IS-2690 (Part II)** | (i) | Water absorption | IS-3495 (Part-II) | 6 tiles out of 12 | 'B' | 216.00 | Samples:- 12 tiles from each source of supply selected at random. |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'C' (Contd…)**

| Sl No | Materials | Test | | Method of testing | Frequency of Tests | Level of test | Rate per test Rs. Ps. | Remarks |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | | 4 | 5 | 6 | 7 | 8 |
| | **Length 150 mm to 250 mm Width 100mm to 200 mm Thickness 35mm to 50mm** | (ii) | Compressive strength | -do- (Part-I) | -do- | 'A' | 180.00 | |
| 8. | **Mangalore pattern roofing tiles** | (i) | Water Absorption | IS-654 (Appx'A') | 6 Tiles out of 32 | 'B' | 180.00 | Samples :- 32 tiles from each consignment of 3000 tiles or part thereof. These tiles shall be checked for dimension. |
| | | (ii) | Breaking Load | -do- | -do- | 'B' | 120.00 | |
| 9. | TIMBER | (i) | Specific gravity and weight | IS-1708- 1960 | Minimum '3' samples from a lot of 4 Cum or 250 pieces of seasoned timber. | 'B' | 120.00 | |
| | | (ii) | Moisture content | -do- | | 'A' | 120.00 | |
| 10. | **Water for construction purposes** | (i) | Test for Acidity | IS-456 & 3015 | Once at the stage of approval of source of water. | 'B' | 240.00 | Also refer clause 4.3 of IS-456 and its subsequent sub clauses regarding suitability of water |
| | | (ii) | Test for Alkalinity | -do- | | 'B' | 240.00 | |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'C' (Contd…)**

| Sl No | Materials | Test | | Method of testing | Frequency of Tests | Level of test | Rate per test Rs. Ps. | Remarks |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | | 4 | 5 | 6 | 7 | 8 |
| | | (iii) | Test for solid content | -do- | | 'C' | 300.00 | |
| 11 | Welding of steel work | | Visual inspection test | IS-822-1970 Clause 7.1 | 100% by visual inspection | Work site | 360.00 | Specialised tests their method and frequency to be decided on consideration of their importance by the Accepting Officer. |
| 12 | Timber panelled and glazed Door/Window& shutters (including Factory made shutters) | (a) | Dimensions, size, workmanship and finish. | IS-1003-1977 (Part-I) | Frequency of sampling form each lot shall be as under :- | | | |
| | | (b) | Strength test: - | | | | | |
| | | (i) | Slamming | IS-1303-1990 | From the each lot 5% of the factory made shutters shall be manufacturer tested for strength tests | | | |
| | | (ii) | Impact indentation | -do- | | | | |
| | | (iii) | Shock resistance | -do- | | | | |
| | | (iv) | Edge loading | -do- | | | | |

| Lot size | Sample |
|---|---|
| 26 to 50 | 5 |
| 51 to 100 | 8 |
| 101 to 150 | 13 |
| 151 to 300 | 20 |
| 301 to 500 | 32 |
| 501 to 1000 | 50 |
| 1001and above | 80 |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'C' (Contd…)**

| Sl No | Materials | Test | | Method of testing | Frequency of Tests | Level of test | Rate per test Rs. Ps. | Remarks |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | | 4 | 5 | 6 | 7 | 8 |
| 13. | **Plywood (IS-303-1989)** | (a) | Moisture content | IS-1734 1983(Part-) | Six tests the pieces cut from each of the boards selected as per table shall be subjected to tests. | 'C' | 240.00 | Sampling shall be as per IS-7835-1975 Tables. |
| 14. | **Wood particle board (medium density IS-3097-1985** | (a) | Density | IS-2360 (Part-III) | Three test specimens from each sample (size 150 mm X 75 mm) | 'A' | 60.00 | Sampling shall be as per IS-3487-83 with moisture meter. |
| | | (b) | Moisture Content | -do- | -do- | 'A'&'B' | 60.00 | |
| | | (c) | Water Absorption | -do- (Part-16) | -do- (Size 300 mm X 300mm) | 'A' | 60.00 | |
| | | (d) | Swelling due to surface Absorption | -do- (Part-17) | -do- (Size 125 mm X 100mm) | 'A' | 60.00 | |
| | | (e) | Swelling in water | -do- | -do- (Size 200 mm X 100mm) | 'A' | 60.00 | |
| | | (f) | Modulus of rupture | -do- (Part-4) | Three test specimens as per IS-2380-77 | 'B' | 90.00 | |
| | | (g) | Screw withdrawal strength | -do- (Part-4) | -do- As per IS-2385 | 'C' | 120.00 | |
| 15 | **Ceramic Tiles** | (a) | Dimensions and surface quality | IS:13630 (Part-1) | As per requirement of IS | 'A' | 10.00 | |
| | | (b) | Modulas of rupture | IS:13630 (Part-6) | As per requirement of IS | 'C' | 100.00 | |
| | | (c) | Moisture expansion using bailing water | IS:13630 (Part-3) | As per requirement of IS | 'A' | 40.00 | |
| | | (d) | Surface abrasions | IS:13630 (Part-11) | As per requirement of IS | 'C' | 100.00 | |

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'D'**

| Ser No | Name of Tests | Frequency | No of test in site lab | No of tests in IITs/SEMT |
|---|---|---|---|---|
| **1** | **CEMENT** | | | |
| **(a)** | **Physical Tests** | Once for each consignment | **100%** | **5%** |
| | (i) Setting time (initial & final) | | | |
| | (ii) Soundness | | | |
| | (iii) Compressive strength | | | |
| | (iv) Fineness | | | |
| (b) | Chemical tests (Chemical composition) | Once for each | 100% in approved NABL accredited lab | 9% |
| 2. | **COARSE AGGREGATE** | | | |
| (a) | Specific gravity | One for each source for sample approval and once more for confirmation in the first lot. | 100% site lab/approved NABL accredited lab | 5% |
| (b) | Water absorption | One for each source for sample approval and once more for confirmation in the first lot. | 100% site lab/approved NABL accredited lab | 5% |
| (c) | Soundness | One for each source for sample approval and once more for confirmation in the first lot. After that once in every month. | 100% site lab/approved NABL accredited lab | 5% |
| (d) | Organic impurities | One for each source for sample approval and once more for confirmation in the first lot. | 100% site lab/approved NABL accredited lab | 5% |
| (e) | Crushing value | One test per each size for every 100 cum | 100% | 5% |
| (f) | Impact value | One test per each size for every 100 cum | 100% | 5% |
| (g) | LosanglesAbrassion test | One test per each size for every 100 cum | 100% | 5% |
| (h) | Deletereous materials | One test per each size for every 100 cum | 100% | 5% |
| (j) | Sieve analysis | One test per each size for every 50 cum | 100% | 5% |
| (k) | Flakiness & elongation index | One test per each size for every 100 cum or part thereof | 100% | 5% |

**PARTICULAR SPECIFICATIONS (CONTD.....)**

**APPX 'D' (Contd….)**

| Ser No | Name of Tests | Frequency | No of test in site lab | No of tests in IITs/SEMT |
|---|---|---|---|---|
| (l) | Moisuture content | Regularly at every one hour interval during batch mixing | 100% | - |
| (m) | Stripping value of aggregates | Initially for aggregate from each quarry and subsequently whenever change in quality of aggregate/quarry noticed. | 100% | 5%(Min once for each quarry) |
| 3. | **FINE AGGREGATE** | | | |
| (a) | Specific gravity | One for each source for sample approval and once more for confirmation in the first lot. | 100% site lab/approved NABL accredited lab | 5% |
| (b) | Organic impurities | One for each source for sample approval and once more for confirmation in the first lot. | 100% site lab/approved NABL accredited lab | 5% |
| (c) | Sieve analysis | One test per each size for every 15 cum | 100% | 5% |
| (d) | Moisture content | Regularly at every one hour interval during batch mixing | 100% | - |
| (e) | Clay and silt impurities | One test per each size for every 15 cum | 100% | 5% |
| **4.** | **WATER** | Initially then quarterly | - | - |
| **5.** | **ADMIXTURE** | Min one test per lot | - | - |
| **17** | **POLYTHENE SHEET** | Only independent test one test per lot | | |

**SIGNATURE OF THE CONTRACTOR**
**DATED:**

**DCWE (CONTRACTS)**
**FOR ACCEPTING OFFICER**